



Supplementary Study Pack for Underperforming Schools

Mathematics JSS 3

March 2021



NOT FOR SALE



TABLE OF CONTENT

TOPIC PAGE	UNIT	
NUMBERS AND	Unit1. Set language and notation.....	2
NUMERATION	Unit2. Basic concept Of sets.....	3
	Unit3. Using different types of numbers to describe sets.....	4
	Unit4. Venn diagram.....	5
	Unit5. Identification of number system (base) used by different cultures.....	7
	Unit6. Conversion of other bases to base10 and from base10 to other bases.....	8
	Unit7. Laws of indices.....	10
	Unit8. Properties of indices.....	12
	Unit9. Standard Forms.....	13
	Unit10. Four operations in mathematics.....	15
	Unit11. Percentages.....	18
	Unit12. Simple Interest.....	22
	Unit13. Compound Interest.....	26
	Unit14. Exchange Rate.....	27
MEASUREMENT AND ESTIMATION	Unit15. Review areas and perimeters of 2D shapes triangles and quadrilaterals.....	31
	Unit16. Areas and circumference of circles.....	37
	Unit17. Volumes and surfaces of 3D shapes: triangular prisms, cuboid and cubes.....	39
GEOMETRY	Unit18. Construction.....	47
	Unit19. Pythagoras.....	51
	Unit20. Congruency.....	56



The Sierra Leone Teaching Service Commission



MONEY AND USE	Unit21. Calculations involving the four operations with money	60
TRIGONOMETRIC		
RATIOS (FRACTIONS)	Unit22. Sine, Cosine and Tangent of angles.....	60
	How to calculate the missing length and angle of a given right-angled triangle	
ALGEBRA	Unit23. Expanding and factorizing brackets.....	62
	Unit24. Linear Equations.....	65
	Unit25. Quadratic Equations	71
	Change of subject and substitution.....	67
	Unit26. Inequality	72
	Unit27 collection and presentation of data	76
STATISTICS AND	Unit28. Averages.....	78
PROBABILITY	Unit29. Probability	81
	Unit30. Internet skills	86
	ANSWERS	89



The Sierra Leone Teaching Service Commission



Dear teacher

These materials are part of an effort taken by the department of the Teaching service Commission (TSC) to improve the quality of teaching and learning of mathematics during the Covid-19 pandemic in Sierra Leone, to serve as supplementary materials for mathematics to the already existing ones. The recourses are designed to make mathematics teaching and learning effective in every community and to ensure continuous improvement in the BECE examinations in Sierra Leone.

As a mathematics teacher, you already know the subject knowledge, therefore your lesson must be approached from a child cantered perspective. You are advised to make explanations, demonstrations and discussions very simple to the level of children's understanding.

Teachers to avoid the use of jargons in the teaching and learning of mathematics and key words must be clearly explained or broken down. Be aware that the teaching and learning must be centered on our young people. Allow children to be actively involved in the teaching and learning process through classroom activities, discussions, demonstrations and carrying out calculations for the entire class. Teachers should give every child an opportunity to access the material in this booklet. This could be achieved through clear explanation and provide enough time for every pupil to complete tasks set in all the units. Teachers to provide an opportunity for pair assessment and self-assessment for a better understanding of every unit completed. All answers for the exercise are set at the back of this booklet.



The Sierra Leone Teaching Service Commission



UNIT 1

NUMBERS AND NUMERATION

SET LANGUAGE AND NOTATION.

TIME: 35 MINS.

TEACHER'S GUIDE

Teachers should use simple language to explain the concepts with clear explanation.

LEARNING OUTCOMES

You can identify elements of a set and use set language and symbols

SET: A set is defined as the collection of well-defined groups of objects called its elements.

{Examples of objects can be numbers, letters, diagrams, symbols etc. The elements of a set are written within curly brackets and are separated by commas. Example $A = \{1, 2, 3, 4, 5\}$

We write $3 \in A$ which means 3 is an element of set A or 3 belongs to set A. $6 \notin A$ which means 6 is not an element of set A. Sets are denoted by capital letters and the elements by small letters. Example. $V = \{a, l, o, u\}$ a set can be described in three ways.

- I. By listing its elements Example. $P = \{2, 4, 6, 8, 10\}$, a set of even numbers from 2 to 10
- II. By word description Example. $P = \{\text{prime numbers from 1 to 10}\}$.

By set builder notation Example $P = \{x: 1 \leq x \leq 10, \text{ where } x \text{ is a prime number}\}$

The number of elements in a set refers to the cardinality of a set Example $S = \{a, e, l, o, u\}$.

The cardinality of set S i.e., $|S| = n(S) = 4$.

Exercise

1. List a set prime number between 10 and 30
2. Write down a set of square numbers between 1 and 40.
3. List down the first 5 (a) square numbers (b) cube numbers
- 4.

UNIT 2

BASIC CONCEPTS and TYPES OF SETS.

TIME: 35 MINS.

TEACHER'S GUIDE

Teachers to clearly explain the meaning the key words like finite sets and to ask students to give their own examples. Teacher to mark and give feedback to students on every exercise completed.

Student Guide

Students to write examples in their exercise books and to ensure exercises set are completed.

LEARNING OUTCOMES



The Sierra Leone Teaching Service Commission



Pupils can apply key words in solving everyday problems involving sets.

A set can be finite or infinite.

FINITE SETS: This refers to a set that has a definite number of elements example. $A = \{1, 2, 3, 4, 5\}$ $n(A) = 5$

INFINITE SET: This refers to the set that has an indefinite numbers of elements example. $R = \{1, 2, 3, 4, \dots\}$.

$R = \{\text{All positives integers}\}$. Here $n(R)$ is unknown the elements are uncountable.

NULL SETS: A set that has no element is called an empty set. It is denoted by $\{ \}$ or $\emptyset$ $\{\phi\}$.

UNIVERSAL SET: A set that has all possible elements within a particular scope. It is denoted by U or E (Eula).

EQUIVALENT SETS: This refers to set that have the same number of elements. Example $P = \{1, 3, 5\}$ and

$Q = \{a, c, e\}$

$n(P) = n(Q)$ therefore P and Q are equivalent.

EQUAL SET: Two sets are said to be equal when they have the same members (elements).

Example $A = \{8, 9, 10\}$ and

$B = \{9, 8, 10\}$

$A = B$ because each member (element) in A has a corresponding member in B . Although the order of listing may not be the same.

SUBSETS: Let A and B be two non-empty sets. If some elements of A are the elements of B , the B is said to be a subset of A denoted by $B \subseteq A$. The symbol means subset and $\supseteq$ means superset.

COMPLIMENT OF A SET. The compliment of a set is the collection of all elements (members) that are in the universal set but not in the set P .

Example $U = \{1, 2, 3, \dots, 10\}$

$P = \{2, 4, 6, 8\}$

The compliment of P denoted by $P^1 = \{1, 3, 5, 7, 9\}$ P and P^1 are disjoint.

UNION AND INTERSECTION OF SET: Let A and B be non-empty sets. The collection of all element that belongs to either A or both A and B refers to the union of A and B .

Example: $A = \{1, 2, 3, 4, 5, 6\}$ and

$B = \{1, 3, 5, 7, 9, 10\}$.



The Sierra Leone Teaching Service Commission



A union B is given by $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 9, 10\}$

8 is not in $A \cup B$ because it is not in A and also not in B.

The collection of all elements that belongs to A and B refers to the intersection of A and B denoted by $A \cap B$. From the above example $A \cap B = \{1, 3, 5\}$ because these elements you can find in both A and B.

Example. IF $U = \{1, 2, 3, \dots, 10\}$

$$A = \{2, 4, 6, 8, 10\}$$

$$B = \{1, 2, 3, 6, 7\}$$

Find the elements of:

- I. A^1
- II. B^1
- III. $A \cup B$
- IV. $A \cap B$
- V. $(A \cup B)^1$
- VI. $A^1 \cap B^1$

Solution

$$\text{Given } U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$A = \{2, 4, 6, 8, 10\}$$

$$B = \{1, 2, 3, 6, 7\}$$

- i. $A^1 = \{1, 3, 5, 7, 9\}$
- ii. $B^1 = \{4, 5, 8, 9, 10\}$
- iii. $A \cup B = \{1, 2, 3, 4, 6, 7, 8, 10\}$
- iv. $A \cap B = \{2, 6\}$
- v. $(A \cup B)^1 = \{5, 9\}$
- vi. $A^1 \cap B^1 = \{5, 9\}$

Exercise

Write down the elements of the following sets:

$$\text{Given } U = \{2, 4, 6, 8, 10\}$$

- I. A^1
- II. B^1
- III. $A \cup B$
- IV. $A \cap B$
- V. $(A \cup B)^1$
- VI. $A^1 \cap B^1$



The Sierra Leone Teaching Service Commission



UNIT 3

Time: 35 mins.

TEACHER'S GUIDE

Teacher to explain the properties of numbers and write them in the form of a set.

STUDENT GUIDE

Pupils to copy the properties of numbers in their exercise books.

USING DIFFERENT TYPES OF NUMBERS TO DESCRIBE SETS.

LEARNING OUTCOMES

Pupils can use different properties of numbers to solve problems with sets

When solving problems under set of members the following description of numbers must be known.

- a) THE SET OF EVEN NUMBER: numbers that can be exactly divided by 2. example {2,4,6,8,10.....}
- b) THE SET OF ODD NUMBERS: Numbers that 2 cannot divided exactly. Example {1,3,5,7,9,11.....}
- c) THE SET OF PRIME NUMBERS: Numbers that have only two factors (i.e. itself and 1) example {2,3,5,7,11,13,17}
- d) THE SET OF WHOLE NUMBERS: Numbers denoted by W. $W = \{0,1,2,3,4,5.....\}$
- e) THE SET OF NATURAL NUMBERS: Numbers denoted by N. Example $N = \{1,2,3,4,5....\}$
- f) THE SET OF INTEGERS: This refers to whole numbers positive and negative. Example $I = \{-3, -2, -1, 0, 1, 2, 3.....\}$
- g) THE SET OF SQUARE NUMBERS. Example {1,4,9,16,25....}.

MULTIPLES OF A NUMBER: Refers to numbers obtained by multiplying n by 1,2,3 etc. example 1 multiples of 5 {5,10,15,20,25....}, example 2. multiples of 3 {3,6,9,12,15,18.....}.

UNIT 4.



VENN DIAGRAMS

Time: 35

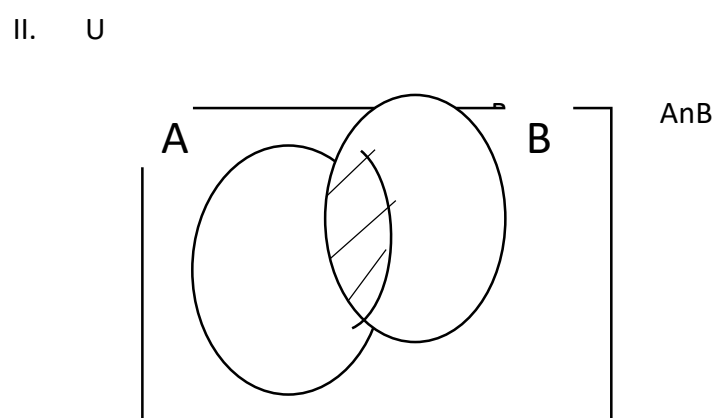
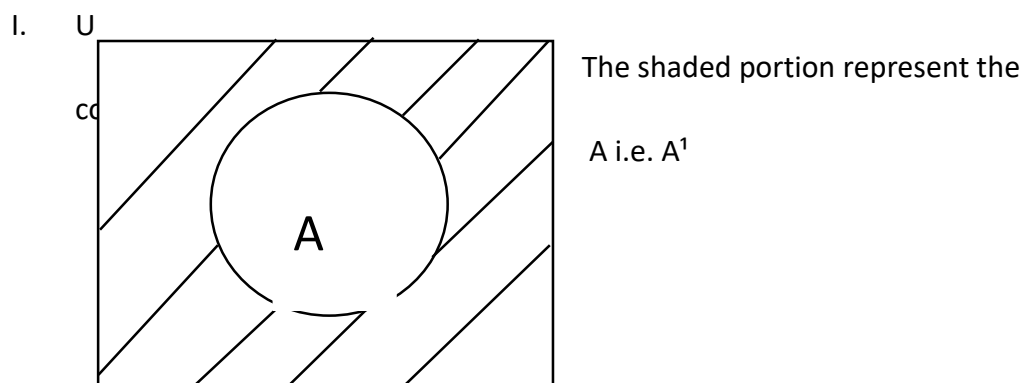
mins.

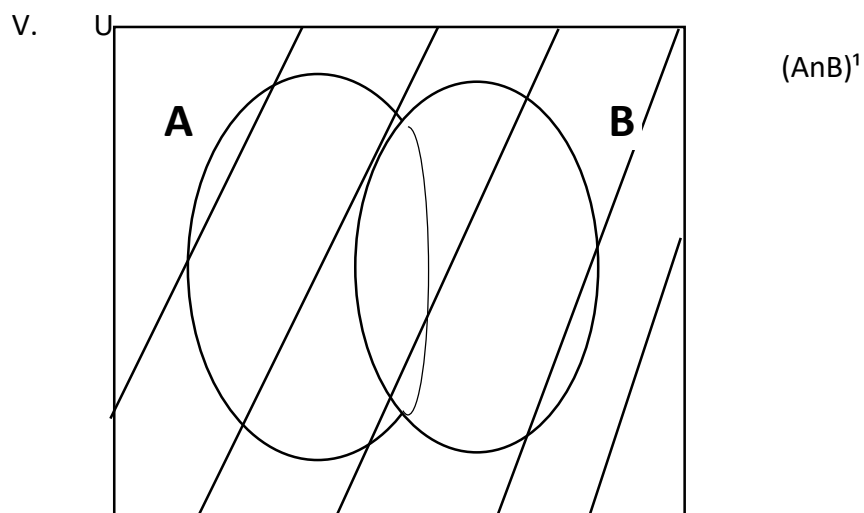
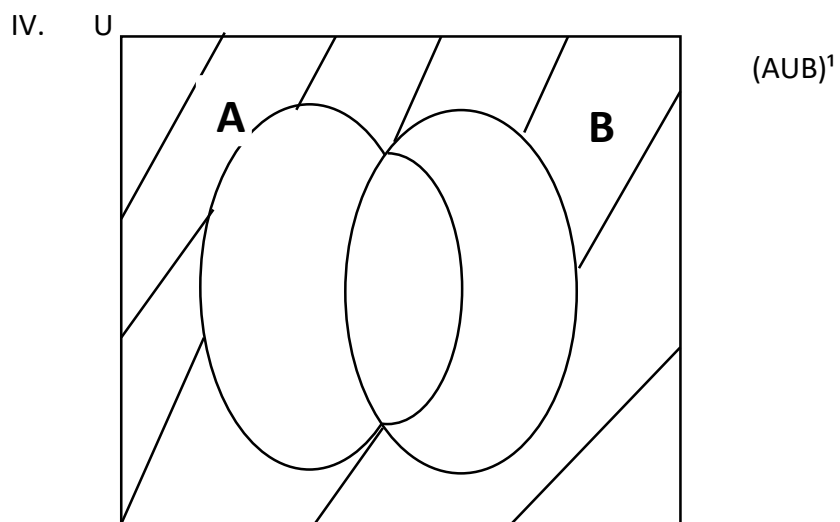
LEARNING OUTCOMES

Pupils can use Venn diagrams to solve simple problems involving sets

This is a rectangle that may contain one, two or three circles that can be used to solve problems in sets. The rectangle always represents the universal set and the circles, the subsets.

SOME RELATIONSHIPS BETWEEN SETS.





Example: given that



The Sierra Leone Teaching Service Commission



$$U = \{1,2,3,\dots,10\}$$

$$P = \{2,4,6,8,10\}$$

$$Q = \{1,2,3,6,7,9\}$$

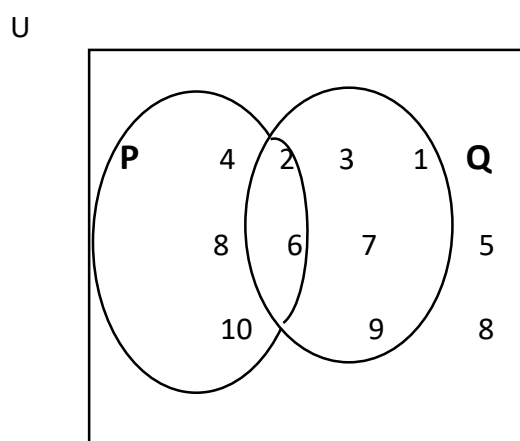
Represent the information on a Venn diagram.

Solution

$$U = \{1,2,3,4,5,6,7,8,9,10\}$$

$$P = \{2,4,6,8,10\}$$

$$Q = \{1,2,3,6,7,9\}$$



$$P \cap Q = \{2,6\} \quad \text{Note that } P \cup Q = \{1,2,3,4,6,7,9,10\}. \quad (P \cup Q)^c = \{5,8\}.$$

Exercise

1. Given that

$$U = \{1,2,3,\dots,10\}$$

$$P = \{2,4,6,8,10\}$$

$$Q = \{1,2,3,6,7,9\}$$

Represent these on a Venn diagram.

UNIT 5.



The Sierra Leone Teaching Service Commission



IDENTIFICATION OF NUMBER SYSTEM (BASE) USED BY DIFFERENT CULTURES

Time: 35 mins.

LEARNING OUTCOMES

TEACHER'S GUIDE

Teacher to explain the different number system used by various cultures.

Pupils can identify number bases used by different cultures

- A number base system is a system for representing numbers, involving a variety of digits or symbols of a given set in a consistent manner. Ideally, a numerical system will:
 - ✓ Represent useful set of numbers (example: all integers or rational numbers).
 - ✓ Give every number represented a unique representation (or at least a standard representation).
 - ✓ Reflect the algebraic and arithmetic structure of numbers.

The ancient Egyptian numbers were used by early Egyptians until early first millennium AD. Some cultures traditionally count in 5's. when counting the days of the week, we count 7's; time in seconds is counted in 60's.

The number system 0,1,2,3,4,5,6,7,8,9 is mostly known as Arabic numbers or Indo – Arabic numbers. The reason why this numerical system is famous, is because it was spread by the Arabs in Europe. The system was developed by the Indian Mathematicians and adopted by

Persian and Arab Mathematicians. This numerical system is the simplest used numerical system

in the world and Sierra Leone and its culture is not an exception. All local languages and culture in Sierra Leone count in Base 10 numerical system I.e., 0,1,2,3,4,5,6,7,8,9.

ROMAN NUMERAL

The symbols are

i.	– 1	X - 10
ii.	– 2	L - 50
iii.	– 3	C - 100
iv.	– 4	D - 500
v.	– 5	M – 1,000



UNIT 6.

NUMBER BASE

Time: 35 mins.

TEACHER'S GUIDE

Teacher should review basic rules in order to write numbers in an index form.

STUDENT'S GUIDE

Students to copy relevant examples in their exercise books.

LEARNING OUTCOMES

Pupils can solve problems involving converting one number base to another number base

HOW TO CONVERT OTHER BASES TO BASE 10.

There are two methods

- 1) The power expansion method and
- 2) The successive multiplication method.

Example. 1

Convert 11001_{two} to base 10.

Power expansion method

$$\begin{aligned} 11001_{\text{two}} &= (1 \times 2^4) + (1 \times 2^3) + (0 \times 2^2) + (0 \times 2^1) + (1 \times 2^0) \\ &= (1 \times 16) + (1 \times 8) + (0 \times 4) + (0 \times 2) + (1 \times 1) \\ &= 16 + 8 + 0 + 0 + 1 \\ &= 25_{10} \end{aligned}$$

Successive multiplication method

$$\begin{aligned} 11001_{(2)} &= (1 \times 2) + 1 \\ &= (3 \times 2) + 0 \\ &= (6 \times 2) + 0 \\ &= (12 \times 2) + 1 \\ &= 25_{10} \end{aligned}$$

Example. 2

Convert 342_{five} to a number in base 10.

Power expansion method

$$\begin{aligned} 342_{(5)} &= (3 \times 5^2) + (4 \times 5^1) + (2 \times 5^0) \\ &= (3 \times 25) + (4 \times 5) + (2 \times 1) \\ &= 75 + 20 + 2 \\ &= 97_{10}. \end{aligned}$$

Successive multiplication method



$$\begin{aligned} 342_{(5)} &= (3 \times 5) + 4 \\ &= (19 \times 5) + 2 \\ &= 95 + 2 \\ &= 97_{10}. \end{aligned}$$

HOW TO CONVERT FROM ONE BASE TO ANOTHER BASE.

Here we first convert to base 10 and by successive division, we convert to the desired base i.e., base 10 is considered as the bridge to the desired destination.

Example. 1

Convert 67_{10} to a number in base 6.

6	67
6	11R1
6	1R5
	0R5

$\uparrow$
 $551_{(6)}$

Since the number was already in base 10, we proceed by successively dividing.

UNIT 7.

Time: 35mins

INDICES

TEACHER'S GUIDE

Teacher to carefully explain the multiplication and division rules with the guided examples.

STUDENT'S GUIDE

Pupils to ensure they copy all the laws of indices and examples in their exercise books.

LEARNING OUTCOMES

Pupils can use multiplication and division rules of indices to solve problems.

LAWS OF INDICES.



The Sierra Leone Teaching Service Commission



$$a^3 = a \times a \times a$$

$$a^4 = a \times a \times a \times a$$

$$\text{So } a^3 \times a^4 = (a \times a \times a) \times (a \times a \times a \times a) \\ = a^7$$

a. The first law of indices known as the multiplication or product rule.

$$a^m \times a^n = a^{m+n}$$

Example. simplify the following:

i. $5^3 \times 5^2$

ii. $2x^4 \times 4x^3$

iii. $3a^2b^{-3} \times 4ab^6$

Solution

$$1. \quad 5^3 \times 5^2 = 5^{3+2} \\ = 5^5 \\ = 3225.$$

$$2. \quad 2x^4 \times 4x^3 \\ = (2 \times 4)^{4+3} \\ = 8^7$$

$$3. \quad 3a^2b^{-3} \times 4ab^6 = (3 \times 4) \times a^{2+1} \times b^{-3+6} \\ = 12a^3b^3$$

Also, if we have $p^4 \div p^3 = \frac{\cancel{p} \times \cancel{p} \times \cancel{p} \times p}{\cancel{p} \times \cancel{p} \times p}$
 $= p$

b. The second law of Indices or the division law

$$a^m \div a^n = a^{m-n}$$

c. The third law of Indices

$$(a^m)^n = a^{m \times n} \\ = a^{mn}$$

Ex. Simplify the following:

i. $21x^{12} \div 14x^3$

ii. $2^5 \div 2^2$

iii. $(4^2 \times 2^3)^2$

iv. $(10b^6c^2)^2$

v. $(12a^4 \times 2a^2) \div (5a^2 \times 6a^3)$

Solution



The Sierra Leone Teaching Service Commission



$$\begin{aligned} \text{i. } 21x^{12} \div 14x^3 &= \frac{21x^{12}}{14x^3} \\ &= \frac{3x^{12-3}}{2} \\ &= \frac{3x^9}{2} \end{aligned}$$

$$\begin{aligned} \text{ii. } 2^5 \div 2^2 &= 2^{5-2} \\ &= 2^3 \\ &= 8 \end{aligned}$$

$$\begin{aligned} \text{iii. } (4^2 \times 2^3)^2 &= 4^{2 \times 2} \times 2^{3 \times 2} \\ &= 4^4 \times 2^6 \\ &= (2^2)^4 \times 2^6 \\ &= 2^{2 \times 4} \times 2^6 \\ &= 2^8 \times 2^6 \\ &= 2^{8+6} \\ &= 2^{14} \\ &= 16384 \end{aligned}$$

$$\begin{aligned} \text{iv. } (10b^6c^2)^2 &= 10^2 \times b^{6 \times 2} \times c^{2 \times 2} \\ &= 100b^{12}c^4 \end{aligned}$$

$$\begin{aligned} \text{v. } (12a^4 \times 2a^2) \div (5a^2 \times 6a^3) &= \frac{12 \times 2 \times a^{4+2}}{5 \times 6 \times a^{2+3}} \\ &= \frac{24a^6}{30a^5} \\ &= \frac{4a^{6-5}}{5} \\ &= \frac{4a}{5} \end{aligned}$$



UNIT 8.

PROPERTIES OF INDICES

Time: 35mins

LEARNING OUTCOMES

Pupils can use use/apply positive and negative fractional indices to solve problems

1. ZERO POWER

$$a^0 = 1$$

2. NEGATIVE INDEX

$$a^{-n} = \frac{1}{a^n}$$

$$a^n$$

3. FRACTIONAL INDEX

$$a^{1/2} = \sqrt{a}$$

$$a^{1/n} = \sqrt[n]{a}$$

Ex. Find the values of

i. $\left(\frac{25}{4}\right)^{1/2}$

ii. $8^{-2/3}$

lii $\frac{2^6 \times 2^2}{4^4}$

iv. $16^4 \times 8^{1/3}$

vi. $27^{-1/3}$

i. $\left(\frac{25}{4}\right)^{1/2} = \frac{25^{1/2}}{4^{1/2}}$

$$4^{1/2}$$

$$= \frac{5}{2}$$

$$2$$

ii. $8^{-2/3} = \frac{1}{8^{2/3}}$

$$8^{2/3}$$

$$= \frac{1}{(2^3)^{2/3}}$$

$$(2^3)^{2/3}$$

$$= \frac{1}{2^{3 \times 2/3}}$$

$$2^{3 \times 2/3}$$



The Sierra Leone Teaching Service Commission



$$= \frac{1}{2^2}$$

$$= \frac{1}{4}$$

$$= \frac{1}{4}$$

$$\begin{aligned} \text{iii. } \frac{2^6 \times 2^2}{4^4} &= \frac{2^{6+2}}{(2^2)^4} \\ &= \frac{2^8}{2^8} \\ &= 2^{8-8} \\ &= 2^0 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{iv. } 27^{-1/3} &= \frac{1}{27^{1/3}} \\ &= \frac{1}{(3^3)^{1/3}} \\ &= \frac{1}{3^{3 \times 1/3}} \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} \text{v. } 16^4 \times 8^{1/3} &= (2^4)^4 \times (2^3)^{1/3} \\ &= 2^{4 \times 4} \times 2^{3 \times 1/3} \\ &= 2^{16} \times 2^1 \\ &= 2^{16+1} \\ &= 2^{17} \end{aligned}$$

UNIT. 9

STANDARD FORM

Time 35 mins.

Pupils to understand that in writing a number in standard form, you can only write one whole number before the decimal point and then write rest of the numbers remaining. Eg $3145 = 3.145 \times 10^3$

pupils to copy examples in their exercise books

LEARNING OUTCOMES

Pupils can use index notation to write large and small numbers using standard forms.

When numbers become very small or large, we can manage them by writing them in standard forms (Radicals). In standard forms we write numbers as:

(a number from / less than 10) x (an integer power of 10).



The Sierra Leone Teaching Service Commission



i.e $X \times 10^n$

where $1 \leq X < 10$ and n is an integer

example. 1 3.82×10^4 is in standard form.

Example .2 25.46×10^5 is not in standard form.

If n is zero, it implies the number is between 1 and 10.

Example. 5.62×10^0

$$= 5.62$$

If n is negative the number may be less than 1.

Example 3.21×10^{-3}

$$= 0.00321$$

If n is positive, it implies the number maybe larger

Example 5.26×10^4

$$= 52600$$

To write a number in standard form we first try to write the number from 1 to less than 10 and then determine the number of the decimal point is being moved.

- If we move the decimal point from right to left, then the power of 10 i.e n is positive
- If we move the decimal point from left to right, then n is negative.

Examples. Write the following numbers in standard forms:

- 5000
- 687.05
- 0.00513
- 0.0246
- 325.68

Solution

- 5000

We write 5.0×10^3 i.e. shifting the decimal 3 spaces from right to left

- $687.05 = 6.8705 \times 10^2$
- $0.00513 = 5.13 \times 10^{-3}$

The decimal point was shifted 3 spaces from left to right, which gives a negative power of 10.

- $0.0246 = 2.46 \times 10^{-2}$
- $325.68 = 3.2568 \times 10^2$



Unit 10.

EVERYDAY ARITHMETIC

Time:35 mins

LEARNING OUTCOMES

Pupils can use/apply BIDMAS to solve word problems.

THE FOUR OPERATIONS

In mathematics, the four operations are ADDITION, SUBTRACTION, MULTIPLICATION AND DIVISION.

When solving problems that involves some or all the operations, we must follow the principles of BODMAS OR BIDMAS as the case may require.

Under this unit attention will be paid to how we use the operations in solving word problems.

Solve: $(5 + 2) \times 3 + 1$ = Solve: $(5 + 2) \times 3 + 1$

Number

BIDMAS is the order in which you work out a calculation

B	I	D	M	A	S
r	n	i	u	d	u
a	d	v	l	d	b
c	i	i	t	i	t
k	c	s	i	t	r
e	e	i	p	i	a
t	s	o	l	o	c
s		n	y	n	t
1st	2nd	3rd	4th	5th	6th



Number

EXAMPLES

1) Work out the following without a calculator:

(a) $2 + 3 \times 4$
 $= 2 + 12$
 $= 14$

(b) $7 - 2 \times 6$
 $= 7 - 12$
 $= -5$

B	O	D	M	A	S
r	r	i	u	d	u
a	d	v	l	d	b
c	e	i	t	i	t
k	r	s	i	t	r
e	2nd	o	p	i	a
t		n	y	o	c
s	1st			n	t
		3rd	4th	5th	6th

(c) $3 + 5 \times 4 \div 2$
 $= 3 + 5 \times 2$
 $= 3 + 10$
 $= 13$

(d) $2 \times 3 - (4 + 2)$
 $= 2 \times 3 - 6$
 $= 6 - 6$
 $= 0$

(e) $4 \times 3^2 - (7 + 2)$
 $= 4 \times 3^2 - 9$
 $= 4 \times 9 - 9$
 $= 36 - 9$
 $= 27$



Work out the answers to these
and then work out the joke:

$$W = 6 + 3 \times 5 \quad R = 3 \times 8 + 2$$

$$G = 3 + 8 \times 2 \quad I = 4 + 7 \times 3$$

$$O = 4 \times 7 + 3 \quad N = 6 + 5 \times 9$$

$$P = 6 \times 5 + 9 \quad T = 14 - 3 \times 2$$

$$S = 16 - 2 \times 7 \quad Q = 7 + 8 \times 2$$

$$U = 5 \times 7 - 8 \quad C = 22 - 3 \times 4$$

$$M = 22 - 3 - 4 \quad L = 28 - 6 \times 2$$

$$D = 7 \times 2 + 4 \quad E = 8 + 7 \times 4$$

$$A = 32 - 6 \times 3 \quad H = 9 \times 3 + 2$$

THE JOKE

21 29 14 8 29 14 39 39 36 51 36 18 8 31

8 29 36 39 16 14 51 8 25 51 8 29 36

15 14 8 29 2 10 16 14 2 2 ?

25 8 19 26 36 21

2 23 27 14 26 36 26 31 31 8 2

SUM AND DIFFERENCE

- The sum of numbers is the result obtained when we add the numbers together.
- Difference refers to the result obtained when we subtract the numbers. It is usual that we always subtract smaller numbers from bigger ones.

Ex.1. The sum of three consecutive numbers is 33. Find the numbers.

Let the 1st number be x

So, the 2nd = x+1

And the 3rd = x+2

Their sum is 33: $x + (x+1) + (x+2) = 33$

$$x + x + 1 + x + 2 = 33$$

$$3x + 3 = 33$$

$$3x = 33 - 3$$

$$\cancel{3}x = \cancel{30}$$

$$\cancel{3} \quad \cancel{3}$$

$$x = 10$$



The Sierra Leone Teaching Service Commission



Ex.2. Find the difference between $33/4$ and $22/3$.

$$\begin{aligned}
 &= 33/4 - 22/3 \quad (\text{we convert to improper and find the L.C.M}) = 13/12 \text{ Or } 11/12 \\
 &= 15/4 - 8/3 \\
 &= \frac{45 - 32}{12}
 \end{aligned}$$

PRODUCT

The product of numbers is the result obtained when the numbers are multiplied.

Ex. The product of two numbers is 24. If one of the numbers is $1/4$, find the other number.

Let the number be y

$$\text{So } y \times 1/4 = 24$$

$$y/4 = 24$$

$$y = 4 \times 24$$

$$y = 96$$

Ex. The sum of 35 and a certain number divided by 4. The result is equal to double the number. Find the number.

Let the number be w

The sum of 35 and a certain number divided by 4:

$$\begin{aligned}
 &\frac{35 + w}{4}
 \end{aligned}$$

The result is double the number:

$$2w$$

$$\begin{aligned}
 \text{So } &\frac{35 + w}{4} = 2w
 \end{aligned}$$

$$(35 + w) = (4 \times 2w)$$

$$35 + w = 8w$$

$$35 = 8w - w$$

$$\begin{aligned}
 &\frac{35}{7} = \frac{7w}{7} \\
 &7 \quad 7
 \end{aligned}$$



The Sierra Leone Teaching Service Commission



$$W = 5$$

Ex. Simplify: $\frac{5 \times 11 \div 3^2}{4}$ (Here the principles of BODMAS or BIDMAS must be applied)

$$\begin{aligned} \frac{5 \times 11 \div 3^2}{4} &= \frac{5 \times 11 \div 9}{4} \\ &= \{(5 \times (11 \div 9))\} \div 4 \\ &= 5 \times 11/9 \times \frac{1}{4} \\ &= 55/36 \text{ or } 119/36 \end{aligned}$$

Ex1. The sum of 8 and a certain number is equal to the product of the number and 3. Find the number.

Ex2. The sum of two numbers is 11. The product of the numbers is 30. Find the numbers.

UNIT II

PERCENTAGES

Time: 70 mins.

LEARNING OUTCOMES

Pupils can solve problems involving profit and loss.

REVISION: A percentage is a fraction in which the denominator is 100. The symbol % means percentage:

Example: 40% mean $\frac{40}{100}$ and 50% means $\frac{50}{100}$

CONVERSIONS

a) **FRACTION TO PERCENTAGE:** To convert fraction to percentage, we multiply 100

E.g., **Convert $\frac{10}{50}$ to percentage**

Solution:

i) $\frac{10}{50} \times \frac{100}{1} = \frac{1000}{50} = 20\%$

ii) $\frac{30}{50}$ to percentage



solution: $\frac{30}{50} \times \frac{100}{1} = \frac{3000}{50} = 60\%$

exercise:

Convert the following fractions to percentage i) $\frac{25}{40}$ ii) $\frac{30}{60}$ iii) $\frac{4}{6}$ iv) $\frac{20}{20}$

b) **TO EXPRESS DECIMALS AS FRACTIONS**

You first convert the decimal to fraction and then multiply your answer by 100.

Examples: Convert (i) 0.62 (ii) 0.683 to percentages

Solution: i) $0.62 = \frac{62}{100} \times \frac{100}{1} = \frac{6200}{100} = 62\%$

ii) $0.683 = \frac{683}{100} \times \frac{100}{1} = 68.3\%$

c) **TO EXPRESS PERCENTAGE AS DECIMAL**

To convert percentage to decimal we divide by 100

Examples: Convert the following percentages to decimals i) 40%
(ii) 1.50% (iii) 25%

Solution: I) $40\% = \frac{40}{100} = 0.40$

ii) $1.50\% = \frac{1.50}{100} = 0.0150$

iii) $25\% = \frac{25}{100} = 0.25\%$

d) **CALCULATING PERCENTAGES OF QUANTITIES**

Examples: i) What is 25% of Le 600?

Solution: Percentage means a quantity out of 100

$25\% = \frac{25}{100}$ and 'of' means multiply by the quantity: $\frac{25}{100} \times \frac{60}{1} = 25 \times 6 =$

Le150

Example:

ii) 20% of a population of 500 people were affected by the COVID 19 disease. How many people were affected?



The Sierra Leone Teaching Service Commission



Solution: $\frac{20}{100} \times 500 = 20 \times 5 =$

Exercise 1:

- a) What is 35% of Le 800?
b) What 25% of a population of 100 people?

e) **DECREASE OF A VALUE BY PERCENTAGE**

Example: Decrease 140 by 10%

Solution:

First method: 140 is 100%, when decreased by 10%, we have 90%

$$90\% \text{ of } 140 = \frac{90}{100} \times 140 = 9 \times 14 = 126$$

Second method:

10% of 140

$$\frac{10}{100} \times 140 = 14$$

$$\begin{aligned} &\text{Reduce 140 by 14} \\ &= 140 - 14 = 126 \end{aligned}$$

Third Method

First find 10% of 140

$$\mathbf{10\% \text{ of } 140 = 140 \div 10 = 14}$$

$$\mathbf{10\% = 14}$$

$$\mathbf{90\% = 14 \times 9 = 126}$$

Exercise 2:

Find the following
without a
calculator:

a) 10% of 80

b) 25% of 60

c) 75% of 800



Ex.3

A builder mold 400 bricks on Saturday and later discovered that 25% of the bricks were faulty. How many bricks were faulty?

PROFIT AND LOSS PERCENTAGES

In the market, the cost price is the price the dealer can buy goods and the selling price is the price at which he sells the goods.

He makes profit when the selling price is more than the cost price. Alternatively, he makes a loss when the selling price is less than the cost price.

PROFIT = Selling Price – Cost Price

LOSS = Cost Price – Selling Price

Profit Percentage = $\text{Actual Profit} \div \text{Actual price} \times \frac{100}{1}$

Loss Percentage = $\frac{\text{Cost Price} - \text{Selling Price}}{\text{Cost Price}} \times \frac{100}{1}$

Examples: (i) A trader bought a plasma television at Le 900,000 and sold it at Le 950,000. Calculate the profit percentage



Solution

$$\text{Profit percentage} = \frac{\text{Selling Price} - \text{Cost Price}}{\text{cost price}} \times \frac{100}{1}$$

$$\frac{950,000 - 900,000}{900,000} \times \frac{100}{1} = \frac{50,000}{900,000} \times \frac{100}{1}$$

$$= \frac{5 \times 100}{90 \times 1} = \frac{5 \times 100}{9} = \frac{50}{9} = \mathbf{55.6\%}$$

Example ii) A chair bought at Le1800 and sold at Le1500. What is the loss percentage?

Solution:

$$\text{Loss Percentage} = \frac{\text{Cost price} - \text{Selling Price}}{\text{Cost Price}} \times \frac{100}{1}$$

$$= \frac{1800 - 1500}{1} \times \frac{100}{1} = \frac{300}{1800} \times \frac{100}{1} = \frac{3}{18} \times \frac{100}{1}$$

$$= \frac{1}{6} \times \frac{100}{1} \times \frac{100}{6} = \mathbf{16.6\%}$$

Example iii) A radio was bought at Le 70,000 and sold at a profit of 20%. Calculate the selling price.

Solution: Profit = $\frac{20\%}{100} \times 70,000$

$$= \mathbf{0.20 \times Le70,000 = Le14,000}$$

$$\text{Selling Price} = \text{Cost Price} + \text{Profit}$$

$$\text{Selling Price is} = \text{Le70,000} + \text{Le14,000} = \text{Le84,000}$$

Exercise 4:

- 1) If 80kg of container is increased by 40%, what is its new value?
- 2) What is 45% of 600
- 3) A dealer sold a textbook for Le800 and made a profit of 25%. At what price did he buy the book?



The Sierra Leone Teaching Service Commission



- 4) A teacher bought a bicycle at Le50,000 and sold it at Le60,000. What is the profit percentage?
- 5) Express 55 out of 75 as a percentage.

UNIT 12:

SIMPLE INTEREST

Time: 70 mins.

LEARNING OUTCOMES

Pupils can calculate interest on problems with loans.

SIMPLE INTEREST: Interest is a payment given for saving or borrowing money. It can also be the price paid for borrowing money. When interest is calculated on the basic sum of money saved or borrowed, it is called SIMPLE INTEREST. Interest is the profit return on investment when money is invested, then interest is paid to the investor.

The money used for the investment is called the **PRINCIPAL**. The percentage return is called the **RATE PERCENT**. The amount is the total formed by adding the interest to the principal.

NOTE: With simple interest, the principal always stays the same, no matter how many years the investment (or loan) may last.

FORMULATE:

Simple Interest can be calculated using the formula

$$I = \frac{PRT}{100} \text{ i.e. } I = \frac{P \times R \times T}{100} \text{ where}$$

P= The Principal (the sum of money saved or borrowed)

R = The Annual rate of interest (Given as a percentage)

T = The Time for which the money is (saved or borrowed)

The other derived formulae are:



The Sierra Leone Teaching Service Commission



$$T = \frac{100 \times I}{P \times R}; R = \frac{100 \times I}{P \times T}; P = \frac{100 \times I}{R \times T}$$

Examples:

- 1) Find the simple interest on Le120 for $7\frac{1}{2}$ years at 60% per annum

Solution: $I = \frac{P \times R \times T}{100}$ Where

I = Simple Interest

R = Rate Percentage

T = Time (in yrs)

$$= \frac{\text{Le}120 \times 6 \times 7\frac{1}{2}}{100}$$

$$= \frac{\text{Le}120 \times 6 \times 15}{100 \times 2} = \underline{\underline{\text{Le}54}}$$

The simple interest therefore is Le54

- 2) Find the simple interest on Le8,000 invested for 3 years at 6% per annum.

$$P = \text{Le}8,000$$

Solution: $I = \frac{P \times R \times T}{100}$ Given

$$R = 6\%$$

- -

$$SI = \frac{8,000 \times 6 \times 3}{100} = 1,280$$

The simple interest = Le1,280

- 3) calculate the rate at which \$4000 invested for 6 years gives an interest of \$4.

$$I = \$4$$

Solution: $R = \frac{100 \times I}{P \times T}$ where

$$P = \$4000$$

$$T = 6 \text{ years}$$

$$: R = \frac{100 \times 4}{4000 \times 6} = \frac{4}{40 \times 6} = \frac{4}{240} = 0.02\%$$



The Sierra Leone Teaching Service Commission



- 4) What is the rate at which Le2000 invested for 4months gives an interest of Le60?

Solution $R = \frac{100 \times I}{P \times T}$

Where **I = LE60, P=**
Le2000

T = 4months

$$: R = \frac{100 \times 60}{2000 \times 0.33} = \frac{6}{2 \times 0.33} = \frac{6}{0.66} = 9.1$$

R= 9.1%

- 5) What sum of money must we invest to give Le300 simple interest at the rate of 60% per annum and the time is 2years?

Solution:

$$SI = \frac{P \times T \times R}{100}$$

Where **R = 6%**

T = 2yrs

I = Le 300

$$P = \frac{100 \times 300}{6 \times 2yrs} = \frac{300000}{12} = \text{Le } 25,000$$

The money invested is Le 25, 000 (Principal)

The amount = Principal + Interest = Le 25000 + Le 300
= Le 25300

Exercise 1

Find the simple interest on the following

- i) Le300 for 4yrs at 8%
- ii) Le520 for 5yrs at 7%
- iii) Le225 for 4yrs at 9%

AMOUNT

The amount is the sum of the principal and the interest i.e. A= Principal + Interest.

Example 1:

Find the simple interest and amount on N120 for 7½ yrs at 6% per annum



The Sierra Leone Teaching Service Commission



$$SI = \frac{PxTxR}{100} = \frac{120 \times 6 \times 7\frac{1}{2}}{100} = \frac{120 \times 6 \times 12}{100 \times 2}$$

$$SI = N54$$

In the example above, the principal is N120 which makes an interest of N54. The Amount now = N120 + N54 = N174

Example 2: Find the amount of Le343.20 in 5yrs at 6½% per annum

Solution:

$$I = \frac{PxRxT}{100} = \frac{Le343.20 \times 6\frac{1}{2} \times 5}{100}$$
$$= \frac{343.20 \times 25 \times 5}{100 \times 4} = \frac{85.8}{4} = \frac{Le 429}{4} = 107.25$$

Amount = Principal+ Interest

$$= Le 343.20 + Le 107.25 = Le 450.45$$

Exercise 2

Find the amount of the following investment

- 1) Le 500 for 1yr at 6%
- 2) N 800 for 1yr at 8%
- 3) \$400 for 3yrs at 6%
- 4) Le700 for 2yrs at 7½%

UNIT 13

COMPOUND INTEREST

Time: 35 mins

Students to copy examples in their books and ensure to remember the appropriate formula for a specific question.

LEARNING OUTCOMES

Pupils can compound interest to calculate total interest over a number of years.



This is the interest accumulated over a certain period

When money is saved with simple interest the interest is paid at regular intervals and the principal remains the same.

With compound interest, the interest is added to the principal at the end of each interval.

Thus, the principal increases and so the interest becomes greater for each interval. Most saving's schemes give compound interest, not simple interest.

Example 1: Find the compound interest on Le600 for 2yrs at 8% per annum

Solution: 1st year: $I = \frac{PxTxR}{100} = \frac{600 \times 8 \times 1}{100}$

$$= \text{Le}48$$

$$\text{Amount at the end of 1st year} = \text{Le } 600 + 48$$

$$= \text{Le}648$$

$$2^{\text{nd}} \text{ year: } I = \frac{PxTxR}{100} = \frac{648 \times 8 \times 1}{100}$$

$$= 6.48 \times 8 \times 1$$

$$= \text{Le}51.84$$

$$\text{Amount at the end of 2nd year} = \text{Le}648 + \text{Le}51.84$$

$$= \text{Le}699.84$$

$$\text{Compound interest} = \text{Le}699.8 - \text{Le } 600 = \text{Le } 99.84$$

Example 2:

Mr. Sankoh borrow Le2000 from Mr. Jajua at the rate of 5% compound interest. How much does Mr. Jajua receive after 4years?



The Sierra Leone Teaching Service Commission



Solution: First year = Principal	Le 2000
Interest = $\frac{5 \times 2000}{100}$	Le 100
2 nd year = Principal	Le 2,100 (2000 + 100)
Interest $\frac{5}{100} \times 2100$	Le 105
3 rd year = Principal	Le 2205
Interest $\frac{5}{100} \times 2205$	Le 110.25
4 th year = Principal	Le 2315.25
Interest = $\frac{5 \times 2315.25}{100}$	Le 115.76
Total amount	= Le 2431.01

Answer: Mr. Jajua receives Le 2431.01 after 4years.

UNIT 14 EXCHNAGE RATES MONEY Time: 35 mins

LEARNING OUTCOMES

Pupils can convert the Sierra Leone LEONES to sterling or dollars.

WEST AFRICAN CURRENCIES

Nigeria	100 Kobo (k)	= 1N Naira
Ghana	100 Pesewas (p)	= 1 Cedi (c)
Sierra Leone	100 cents (c)	= 1 Leone (Le)
The Gambia	100 bututs (b)	= 1 Dalasi (D)
Liberia	100 cents (c)	= 1 Dollar (\$)
French speaking countries – France CFA (undivided)		



Other Countries

Britain 100 pence (p) = 1 pound (£)

USA 100 cents (c) = 1 Dollar (\$)

EXCHNAGE RATE

The exchange rates fluctuate i.e., rises or fall. At the time of going to press £1 sterling was equivalent to the following

- The Gambia D 4.00
- Ghana c 4.85
- Liberia \$ 1.77
- Nigeria N 1.18
- Sierra Leone Le 2.18

French speaking countries = 592 for CFA

The following table gives the exchange rates between the English-speaking countries of West Africa.

			\$		Le
D1	1.00	1.21	0.44	0.30	0.55
C1	0.82	1.00	0.36	0.24	0.45
\$1	2.26	2.74	1.00	0.67	1.25
N1	3.39	4.11	1.50	1.00	1.85
Le1	1.83	2.22	0.81	0.54	1.00

Note: As already mentioned exchange rates changes from day to day. The above rates may only be taken as approximations for practice purposes.



COPY INTO YOUR BOOKS

Country	Exchange rate for £5 sterling
Australia	2.82 Australian dollars
European countries	1.58 euros
Switzerland	2.32 Swiss francs
South Africa	16.34 rand
Turkey	2520000 New Turkish lira
U.S.A.	1.55 US dollars

CHANGING TO A FOREIGN CURRENCY

FOREIGN CURRENCY = HOME CURRENCY x EXCHANGE RATE

Amadu is going to Spain on a holiday to watch Real Madrid FC and changes £200 to euros. How many euros does he get?

STEP 1: FIND THE EXCHANGE RATE IN THE TABLE

STEP 2: INSERT AMOUNTS INTO THE FORMULA

FOREIGN CURRENCY = BRITISH CURRENCY x EXCHANGE RATE

AMOUNT OF EUROS = £200 x 1.58

AMOUNT OF EUROS = € 316

£200 = €316



The Sierra Leone Teaching Service Commission



Change £150 into Australian Dollars

£150 x 2.82

\$ 423

Change £390 into US Dollars

£390 x 1.55

\$ 604.50

Change £250 into Euros

£250 x 1.58

€ 395

Change £1500 into rand

£1500 x 16.34

24510 rand

Change £500 into Swiss Francs

£500 x 2.32

**1160
Swiss
Francs**

Exercise

Exchange Rates (£1 =)

QUESTIONS

Europe	1.15 euros	Japan	125.97 yen
Australia	1.57 Dollars	India	57.23 Rupees
USA	1.52 Dollars	S.Africa	10.63 Rand

Change £450 TO USA Dollars	Change £700 to Rupees
Change £620 to Yen	Change £500 to Rand
Change £350 to Euros	Change £250 to Australian Dollars



MEASUREMENT AND ESTIMATION

Time: 70 mins

UNIT 15. REVIEW OF AREAS AND PERIMETERS OF TRIANGLES AND QUADRILATERALS

LEARNING OUTCOMES

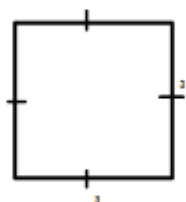
Pupils can recall and use the formulas for the area and perimeter of 2D shapes.

A polygon of four sides is called a quadrilateral e.g. (i) square (ii) rectangle (iii) Rhombus

(iv) Trapezium (v) Parallelogram

Examples

a) Square



All sides are equal

$$\text{Area (A)} = a^2 \text{ (a x a)}$$

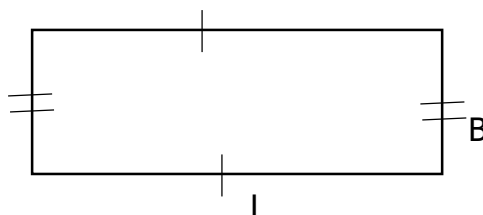
$$\text{Perimeter} = P = a + a + a$$

PERIMETER:

The word comes from two Greek words "Per" meaning round and meter meaning to measure.

Therefore, perimeter means "to measure round". The distance round the edge of a shape.

b) RECTANGLE



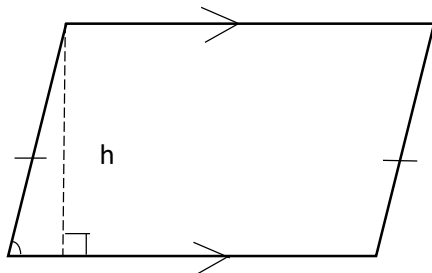
Two opposite sides are equal



$$\text{Area (A)} = LB = L \times B$$

$$\text{Perimeter (P)} = 2 (L+B)$$

c) PARALLELOGRAM

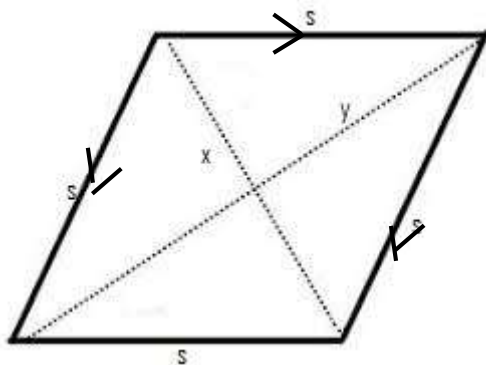


Opposite sides are equal and parallel

$$\text{Area (A)} = Bh$$

$$\text{Perimeter (P)} = 2(a+b)$$

d) RHOMBUS

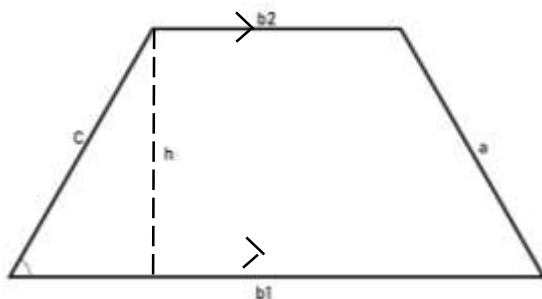


All four sides are equal, and the opposite sides are parallel

$$\text{Area(A)} = \frac{1}{2}xy$$

$$\text{Perimeter(P)} = 4s$$

e) TRAPEZIUM



Has one parallel side.

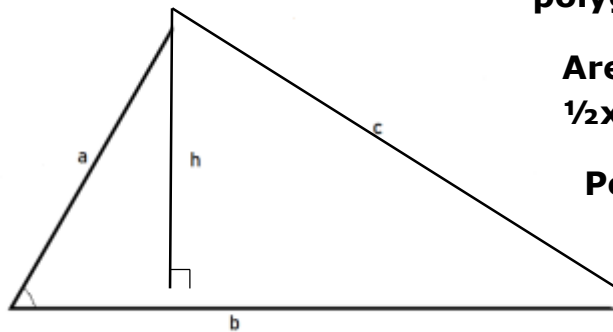
$$\text{Area (A)} = \frac{1}{2}(b_1 + b_2)h$$

$$\text{Perimeter (P)} = a+b_1+b_2+c$$



f) **TRIANGLE**

A shape with three sides or a polygon with 3 sides.

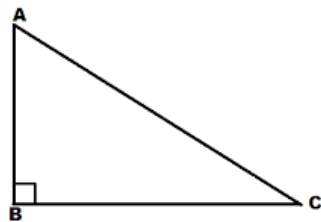


$$\text{Area}(A) = \frac{1}{2}\text{base}(b) \times h \text{ or } \frac{1}{2}bh \text{ or } \frac{1}{2}bxh$$

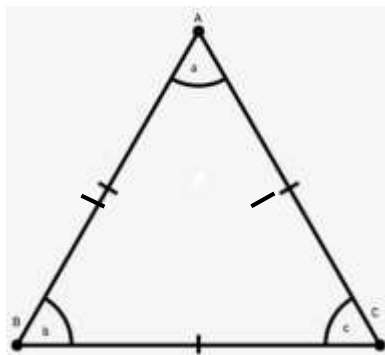
$$\text{Perimeter} = a + b + c$$

TYPES OF TRIANGLES

- i) **A Right-Angle Triangle:** The longest side is opposite to the right angle and is 90°

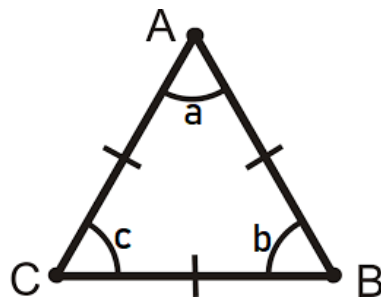


- ii) **An Isosceles Triangle:** Two sides of this triangle are the same and the base angles are equal.



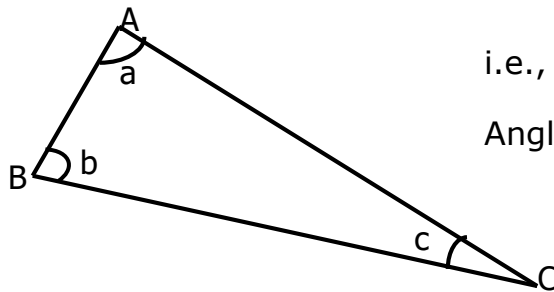
i.e., Side $AC = AB = BC$ and angle $b = C = a$

- iii) **Equilateral Triangle:** All the three sides and angles are equal. Hence all the angles are equal



i.e., Side $AC = AB = BC$ also $a = b = c$

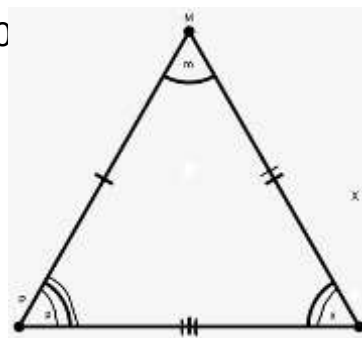
- iv) **Scalene Triangle:** All the three sides are different; so are its angles.



i.e., Side $AB \neq BC \neq AC$

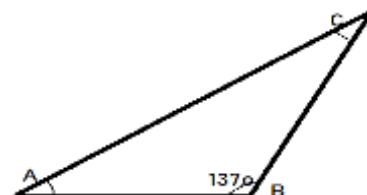
Angle $a \neq b \neq c$

- v) **Acute Angle Triangle:** Each angle for this triangle is less than 90°



i.e., Side $PM = PX = MX$ and angle $p = x = m$

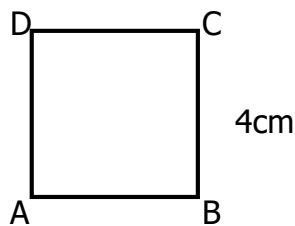
- vi) **An Obtuse Angle Triangle:** One of the angles is greater than 90°





CALCULATIONS: Calculate the area of the following figures

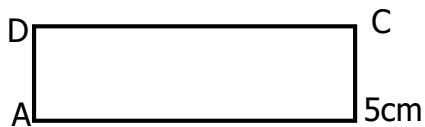
EXAMPLE (1)



$$\text{Area} = L \times B = 4\text{cm} \times 4\text{cm} = 16\text{cm}^2$$

$$\text{Perimeter} = 4\text{cm} + 4\text{cm} + 4\text{cm} + 4\text{cm} = 16\text{cm}$$

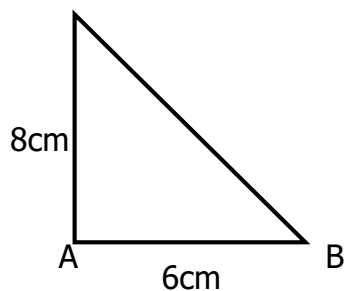
EXAMPLE (2)



$$\text{Area} = L \times B = 6\text{cm} \times 4\text{cm} = 24\text{cm}^2$$

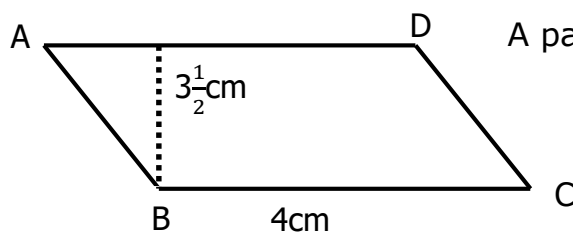
$$\text{Perimeter} = 6\text{cm} + 4\text{cm} + 6\text{cm} + 4\text{cm} = 20\text{cm}$$

EXAMPLE (3)



$$\text{Area} = \frac{1}{2}bh = \frac{1}{2} \times 6 \times 8 = 24\text{cm}^2$$

EXAMPLE (4)



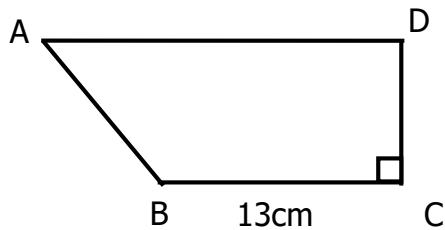
A parallelogram Area = bh

$$b = 4\text{cm}$$

$$h = 3\frac{1}{2}\text{cm}$$

EXAMPLE (5)

Calculate the area of the figure below

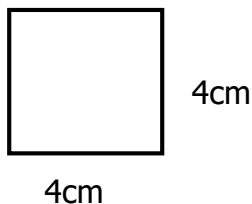


The figure is a Trapezium DC is I to BC. Therefore, DC is the distance between AD and BC Area of Trapezium =

$$\frac{1}{2} (15+13) \times 6 = \frac{1}{2} (28) \times 6$$
$$= 14 \times 6 = 84 \text{cm}^2$$

EXAMPLE (6)

Find the Perimeter of a square with sides 4cm.

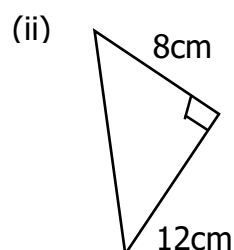
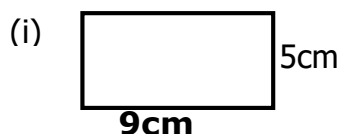


$$P = 4 + 4 + 4 + 4 = 16 \text{cm}$$

$$\text{Or } 2(4+4) = 2(8) = 16 \text{cm}$$

Exercises 1

- 1) The area of a rectangle is 220cm^2 . If the width is 25cm. find its length
- 2) What is the area of a parallelogram whose base is 7cm long and its vertical height is 4cm?
- 3) Find the diameter of a circle whose circumferences are (i) 34.4m (ii) 18.5cm
- 4) Find the circumferences of a circle whose radius is 6cm?
- 5) If the area of circle is 144cm^2 . Find its radius
- 6) Calculate the area of circle of the following figures

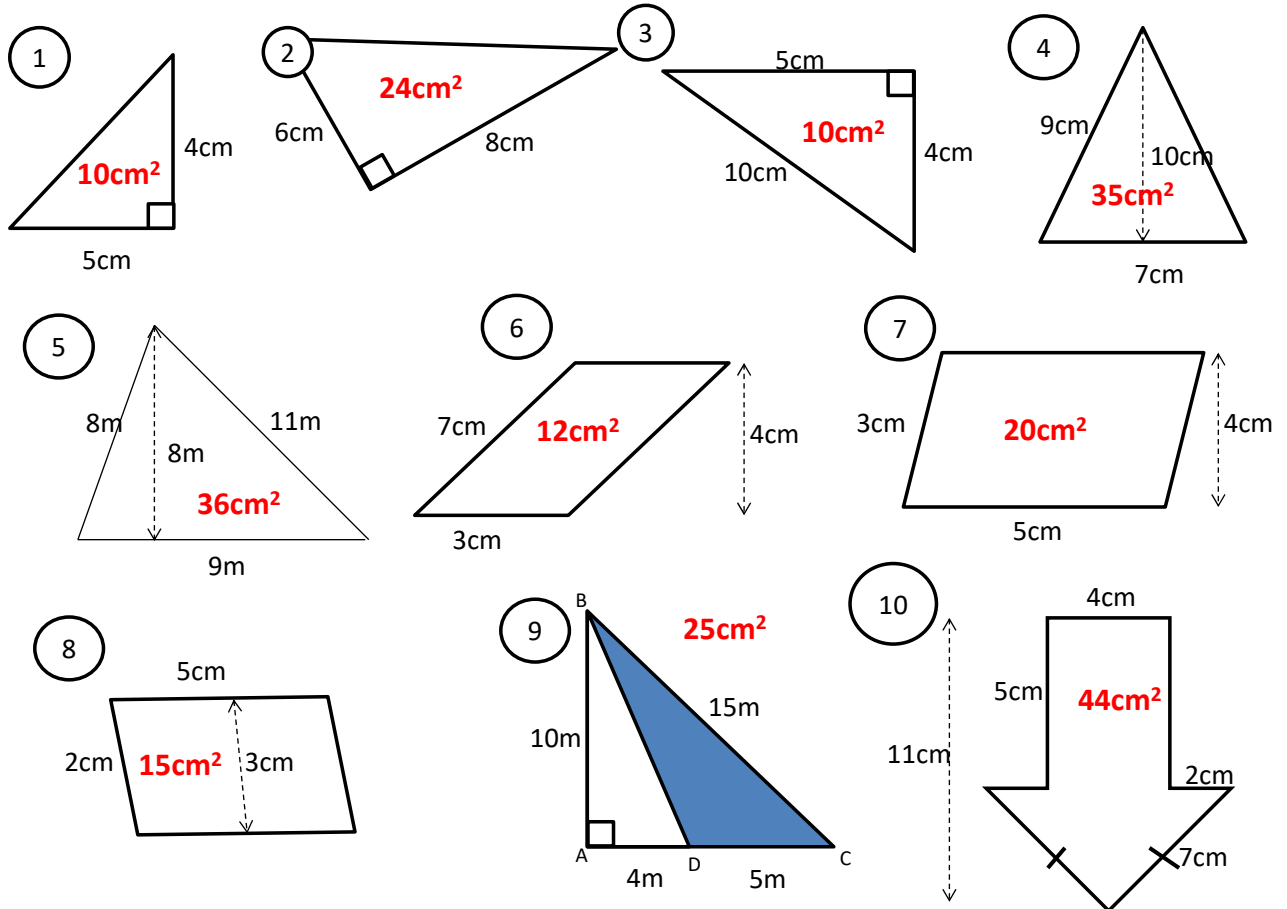




7. The area of a triangle is 76.85cm^2 . If its height is 10.6cm . what is the length of the base of the triangle?

Exercise 2

Calculate the area of the following shapes. In Q9, calculate the shaded area.



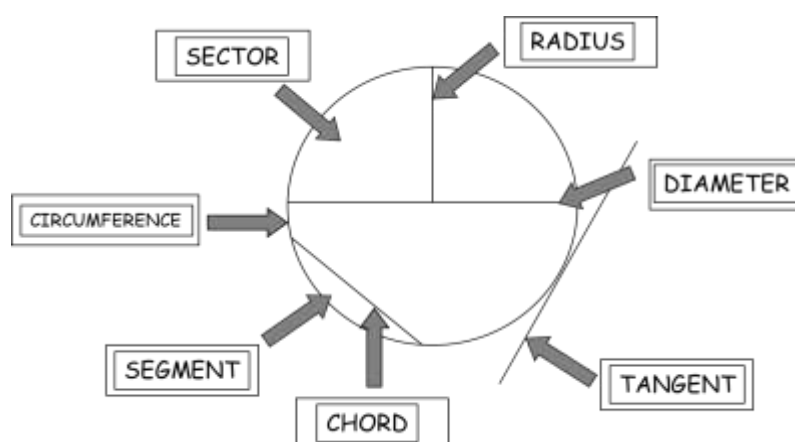


UNIT 16

CIRCLE Time: 70 mins.

LEARNING OUTCOMES

Pupils can use formulas to calculate areas and perimeters/circumference of circles.



Note: π is an appropriation for a special number represented by the Greek Letter π (Pi).

For any circle $\pi = \text{circumference} \div \text{diameter}$

The approximation value is $\frac{22}{7}$. Alternatively, we sometimes use 3.142.

$$C = \pi d.$$

The Length of the diameter doubles the radius,
so, $C = 2\pi r$.

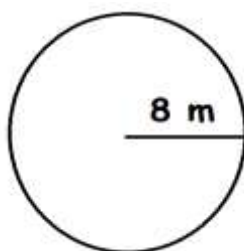
Example 1: Find the area of a circle of radius 8cm.

Solution:

$$\text{Area} = \pi r^2$$

$$A = \frac{22}{7} \times 8^2 = \frac{22}{7} \times 8 \times 8$$

$$= \frac{22}{7} \times 64 = \frac{1408}{7}$$





Area = 201cm

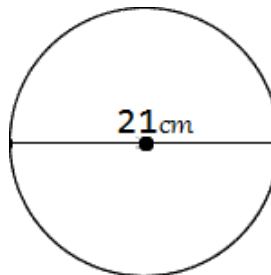
Example 2: Find the Circumference of a circle with diameter 21cm.

Using $\pi = \frac{22}{7}$.

$Cir = \pi d$

$$C = \frac{22}{7} \times 21 = \frac{462}{7}$$

Circumference = 66cm



Example 3: The radius of a circle is 7cm.

(a) What is the length of its diameter?

(b) What is the length of its circumference?

Solution: (a) Diameter = 2 x radius

Radius = 7cm

Diameter = 2 x 7 = 14cm

(b) Circumference = $2\pi r = 2 \times \frac{22}{7} \times 7$

$$\frac{44}{7} \times 7 = 44\text{cm}$$

Example 4:

The circumference of a circle is 176cm.

Find (a) Radius (b) Diameter

Solution: (a) Circumference/Perimeter = $2\pi r$

$$\therefore 176\text{cm} = 2 \times \frac{22}{7} \times r$$

$$176 \times 7 = 2 \times 22 \times r$$

$$\frac{176 \times 7}{2 \times 22} = r$$

$$\therefore r = \frac{1232}{44} = 28\text{cm}$$

Therefore, the radius = 28cm

(b) Diameter = 2r

Where r = 28cm

$$\therefore 2 \times 28 = 56\text{cm}$$



Example 5:

What is the radius of a circle with area 616cm^2 ? $\pi = \frac{22}{7}$

Solution:

$$\text{Area} = \pi r^2$$

$$\therefore 616 = \frac{22}{7} \times r \times r$$

$$616 \times \frac{7}{22} = r \times r$$

$$r = \sqrt{(28 \times 7)} = \sqrt{(4 \times 7 \times 7)}$$

$$r = \sqrt{(2 \times 2 \times 7 \times 7)}\text{cm}$$

$$r = 2 \times 7$$

$$\therefore \text{radius} = 14\text{cm}$$

Exercise

Calculate the area of the following circles:

1. Radius is 3 cm.
2. Diameter 10 cm
3. Radius 7 cm

UNIT 17

Time: 70 mins.

VOLUME AND SURFACE OF THREE DIMENSIONAL SHAPES

LEARNING OUTCOMES

Pupils can apply formulas to calculate total surface areas and volumes of 3D shapes.

Volume is the measure of space occupied by a solid. Volume can be equated to the amount of material used to make the solid if the solid is not hollowed.

Capacity is the measure of the amount of content a container can hold. If the container is made of thin metal, its volume is equivalent to its capacity.



Volume is measured in cubic units. If the metric system Litre is the unit used for measuring capacity and pints, quarts and gallons are units in the imperial system used for measuring capacity.

The cross-section of a solid shape is the shape found when the solid is cut through parallel to its end face.

Prism: Any solid with a constant cross-section is a prism.

A Solid Cylinder: Is a circular prism whose cross-section is a circle.

Therefore:

Volume of a Circular Prism = Volume of Cylinder

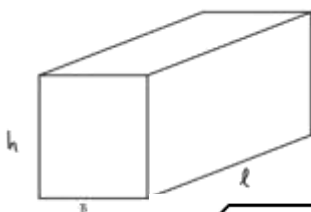
= Area of circular end face x Height

= Area of cross-section x Height

A cuboid is a rectangular prism whose cross-section is a rectangle.

Note: The volume of a prism is the product of the area of its cross-section and its height.

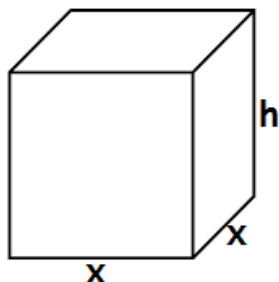
(a) Cuboid (Rectangular Tank)



$$\begin{aligned}\text{Area} &= 2Lb + 2Lh + bh \\ &= 2(Lb + Lh + bh)\end{aligned}$$

$$\text{Volume} = L \times b \times h$$

(b) Square B

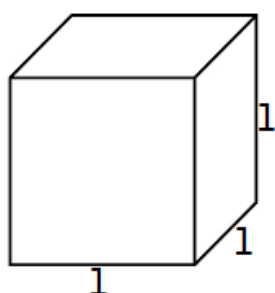


$$\text{Area} = 2x^2 + 4 \times h;$$

When top opens

$$\text{Area} = x^2 + 4 \times h$$

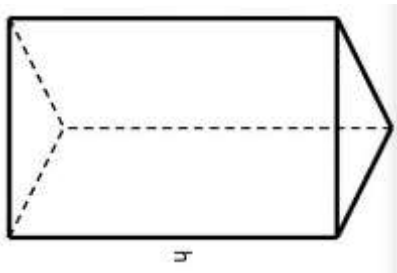
(c)



$$\text{Area} = 6L^2; \text{ when top opens}$$

$$\text{Area} = 5L^2$$

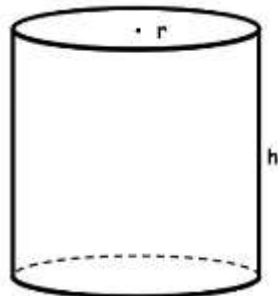
(d) Prism



Curved Surface Area i.e.
Perimeter of Cross-section $\times$
Length of Solid + (Total Area of
Solid)

$$V = Ah \text{ (AL) Where } A = \text{Area}$$

(e) Cylinder (Milk Tin)



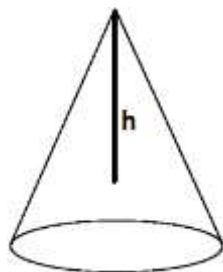
Curved Surface Area

$$A = 2\pi r(r + h)$$

When top opens

$$A = \pi r^2 + 2\pi L = \pi r(r + 2h)$$

(f) Cone (funnel)

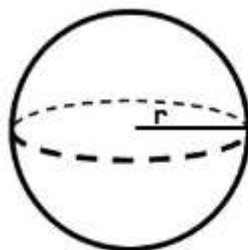


Curved Surface Area

$$A = \pi r^2 + 2\pi rL = \pi r(r + L)$$

$$V = \frac{1}{3} \pi r^2 h$$

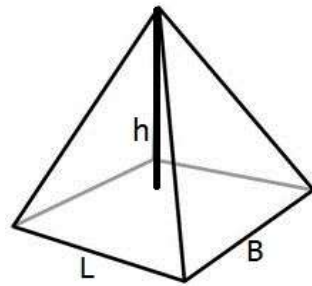
(g) Sphere (Earth)



Surface Area $A = 4\pi r^2$

$$V = \frac{4}{3} \pi r^3$$

(h) Pyramid

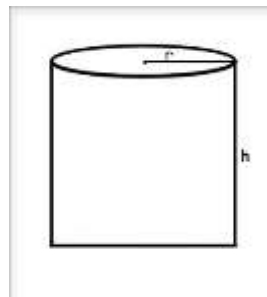


Sum of the area of the triangle forming the side plus the area of the base

$$V = \frac{1}{3} Lbh \text{ i.e.}$$

$$V = \frac{1}{2} \times L \times b \times h$$

(i) Hemisphere



Surface Area

$$A = 2\pi r^2 \text{ and } V = \frac{2}{3}\pi r^3$$

Area and Volume

LO: Calculate the volume of cuboids and cylinders.



$$\text{Area} = \quad \times$$

$$\begin{aligned} \text{Volume of a} \\ \text{Cylinder} \\ = \\ \text{Area of Circle} \times \\ \text{Height} \end{aligned}$$

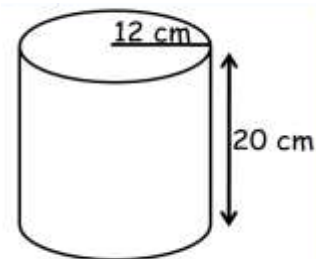
Area and Volume

LO: Calculate the volume of cuboids and cylinders.

$$\text{Volume of a prism} = \text{area of cross section} \times \text{length}$$

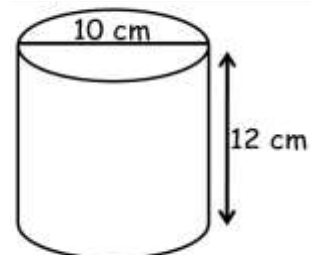
Calculate the volume of this cylinder. Give your answer to 1 d.p.

$$\begin{aligned} \text{Area of cross-section} &= \pi \times 12^2 \\ &= 452.389... \text{ cm}^2 \\ \text{Volume} &= 452.389... \times 20 \\ &= 9047.8 \text{ cm}^3 \end{aligned}$$



Calculate the volume of this cylinder. Give your answer to 3 s.f.

$$\begin{aligned} \text{Area of cross-section} &= \pi \times 5^2 \\ &= 78.539... \text{ cm}^2 \\ \text{Volume} &= 78.539... \times 12 \\ &= 943 \text{ cm}^3 \end{aligned}$$





Examples

- 1) A rectangular Tank is 76cm long, 50cm wide and 40cm high. How many liters of water can it hold?

Solution: $V = Lbh = 76\text{cm} \times 50\text{cm} \times 40\text{cm}$

$$V = 152000\text{cm}^3$$

$$1000\text{cm}^3 = 1 \text{ Litres}$$

$$\therefore 152000\text{cm}^3 = \frac{152000}{1000} = 152 \text{ Litres}$$

Prism – A shape with uniform cross-section.

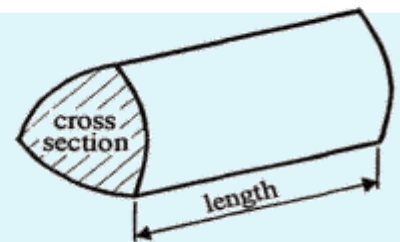
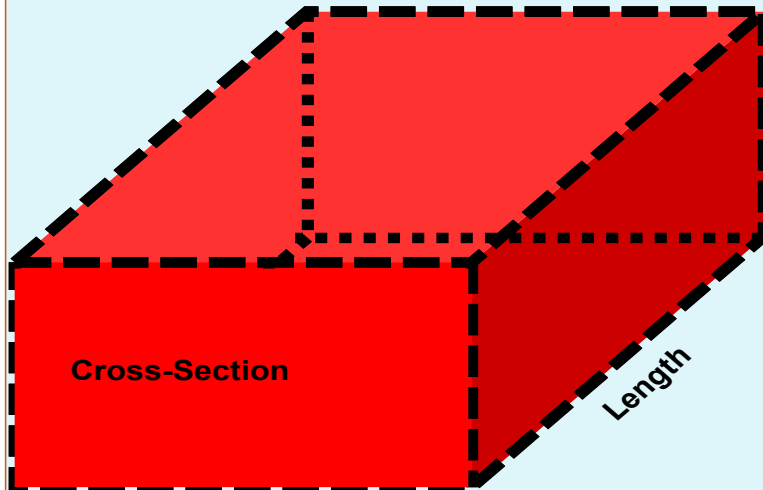
If you cut off a slice from a prism, its volume is the same across the whole shape.

Volume of a prism

Volume



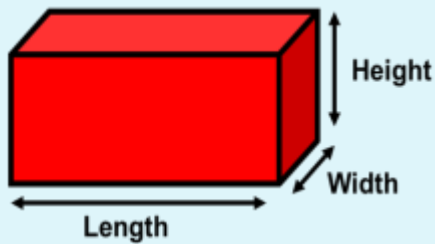
Volume of prism = area of cross section $\times$ length



**VOLUME IS THE SPACE
INSIDE A 3D SHAPE**

Volume of cuboid = Area of rectangle **x Length**

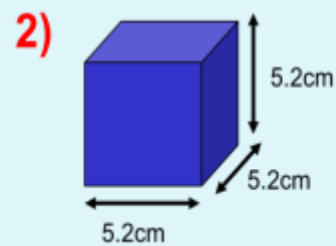
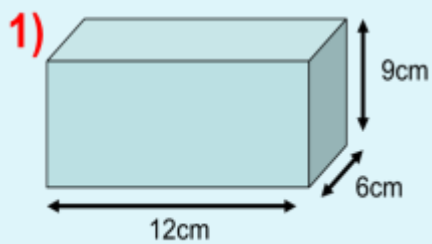
Volume



Volume of a cuboid = $L \times W \times H$

EXAMPLES

Find the volume of the cuboids below in cm^3 :



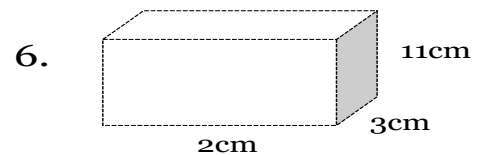
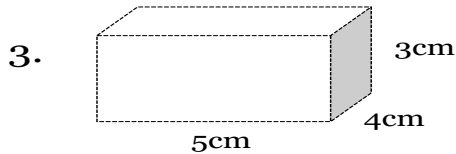
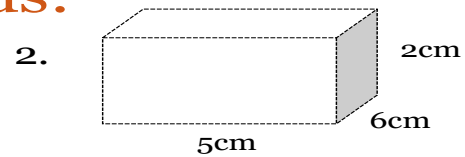
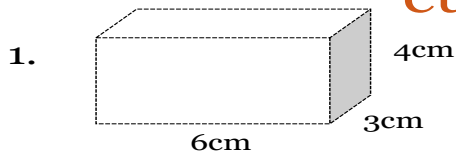
Answers

1) Volume = $12 \times 6 \times 9 = 648 \text{ cm}^3$

2) Volume = $5.2 \times 5.2 \times 5.2 = 140.608 = 140.6 \text{ cm}^3$



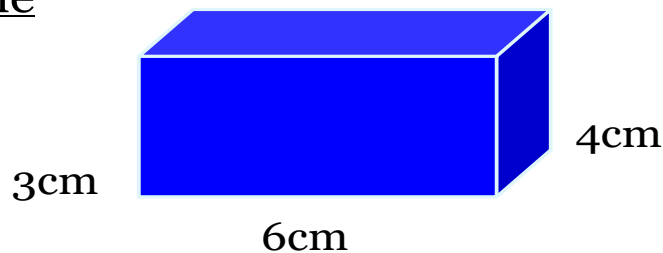
Find the volume of the following cuboids:



Calculating a Missing Length with Given Volume

Sometimes in an exam, you will be asked to find a missing length of a cuboid with a given volume.

Example



$$\text{Volume} = 72\text{cm}^3$$

$$3 \times 4 = 12$$

$$72 \div 12 = 6$$

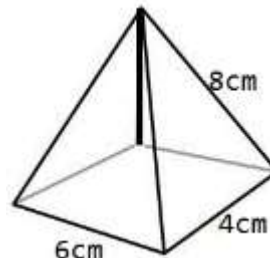
Step 1 - multiply the two given values.

Step 2 - divide the volume by the answer you got when you multiplied the two values.



Pyramids

A Pyramid 8cm high stands on a rectangular base 6cm by 4cm. Calculate the volume of the pyramid.



$$\text{Volume} = \frac{1}{3} Lbh$$

$$V = \frac{1}{3} \times 6 \times 4 \times h = 64\text{cm}^3$$

Exercises

- 1) Find the volume of a circular prism 10ft long if the radius of the cross-section is 7ft. Use $\pi = 3\frac{1}{7}$
- 2) A triangular prism is 20cm long. The triangular face has a base 8cm long and the perpendicular height to the base is 7cm. what is the volume of the prism?
- 3) Find the height in meters of a pole if the volume of wood used to make the pole is 200cm^3 and the area of the cross-section is 154cm^2

UNIT 18

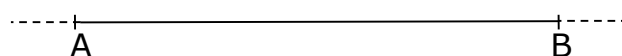
GEOMETRICAL CONSTRUCTIONS

Time:70 mins

LEARNING OUTCOMES

Pupils can bisect lines and angles.

A) To Bisect a Straight-Line Segment

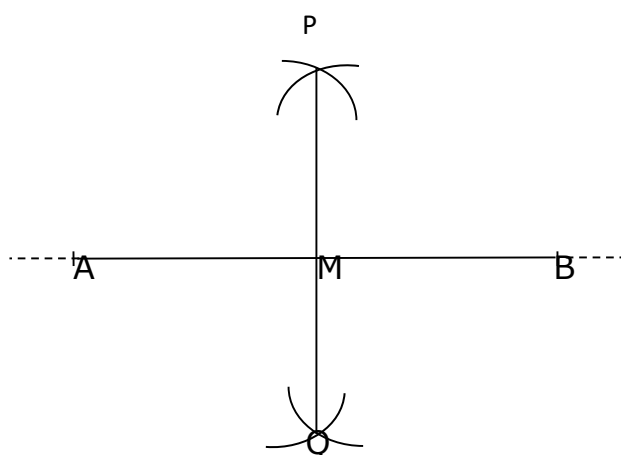


The line segment AB is the part of the line between A and B including the points A and B.



To bisect the line segment AB means to divide it into two equal parts.

- i) Draw a straight line through P and Q so that it cuts AB at M.



M is the midpoint of AB. PQ meets AB perpendicularly. PQ is the perpendicular bisector of AB. Use a ruler and protractor to check that $AM = MB$ and $\angle AMP = \angle BMP = 90^\circ$



Constructions

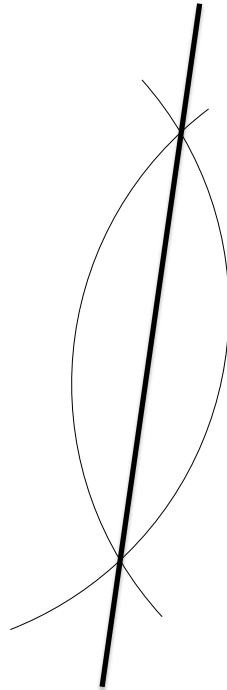
LO: Construct bisectors, angles and loci.

Draw any two points, label them A and B, and find their **perpendicular bisector**.

A ●

STEP 1:

- Compass on A, set the distance slightly more than halfway between A and B.
- Draw an arc.



STEP 2:

Using the same distance on your compass, draw another arc.

● B

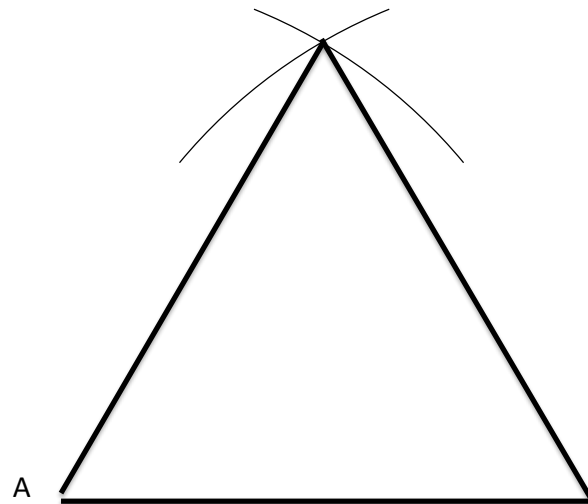
STEP 3:

Draw a line between the two points of intersection.

Constructions

a. Equilateral Triangle

Draw a line of suitable length (e.g. 7cm) in your books, leaving some space above.
Construct an equilateral triangle with base AB.

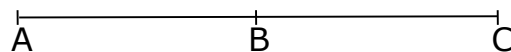


Method

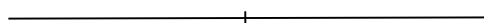
1. Draw two arcs with the same length AB, with centres A and B.
2. Join points A and B to the arc to form the equilateral triangle.

B) To Construct an Angle of 90°

Given a point on a straight-line AC



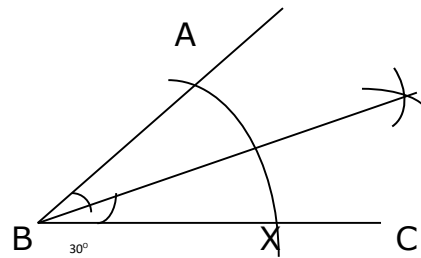
You are required to construct a line BR through B such that $\angle RBA = \angle RBC = 90^\circ$



C) To Construct an Angle of 30°



30° is $\frac{1}{2}$ of 60° . To construct an angle of 30° , first construct an angle of 60° and then bisect it as shown below.



Use a protractor to check the data above.

Exercises: Use ruler and a pair of compasses only in this exercise

- 1) Draw any angle $\angle ABC$. Use the above method to construct the bisector of $\angle ABC$. Use a protractor to check your result.
- 2) Construct angles of 90° and 45° .



UNIT 19. PYTHAGORAS THEOREM Time: 70 mins

TEACHER'S GUIDE

Teachers to carefully review area of a square which is related to Pythagoras theorem. To understand that finding the hypotenuse, you add the two lengths. But finding any of the short sides, you subtract.

STUDENT GUIDE

Students to learn how to draw a right-angle triangle. To calculate the hypotenuse, you add two lengths that are squared. To find any short side, hypotenuse squared minus the square of the short side.

LEARNING OUTCOMES

Pupils can use Pythagoras theorem to solve problems.

Pythagoras' Theorem

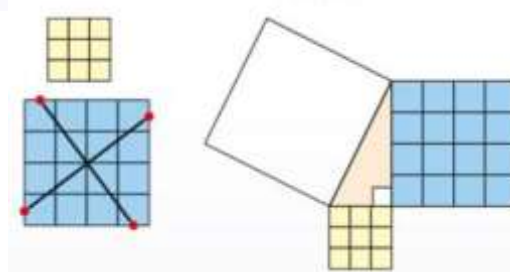
LO: Use the theorem to find a missing side in a right angled triangle.

Draw a 3cm, 4cm, 5cm right-angled triangle with a square on each side.

Cut out another 3cm by 3cm square.

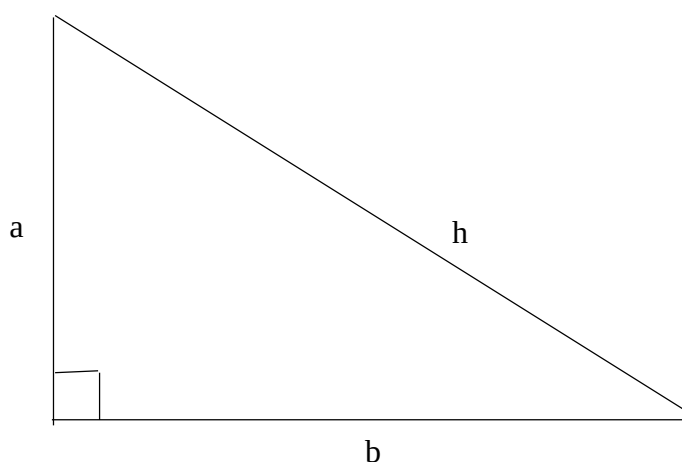
Cut out another 4cm by 4cm square and cut it into 4 pieces exactly as shown.

Use these five pieces to fit in the 5cm by 5cm square to demonstrate Pythagoras' theorem.





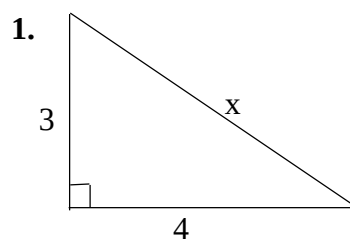
In a right-angled triangle the longest side opposite the right angle is called the HYPOTENUSE.



The side h being the longest is called the hypotenuse side. If “ a ” and “ b ” are the lengths of the sides which make up the right angle, then $h^2 = a^2 + b^2$.

The theorem therefore states that, “In a right-angled triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides forming the right angle”.

Find the values of the sides marked with letters.



Solution

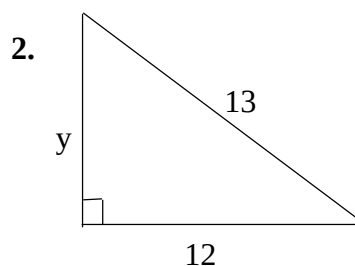
$$x^2 = 3^2 + 4^2$$

$$x^2 = 9 + 16$$

$$x^2 = 25$$

$$x = \sqrt{25}$$

$$x = \underline{5}$$



Solution

$$13^2 = y^2 + 12^2$$

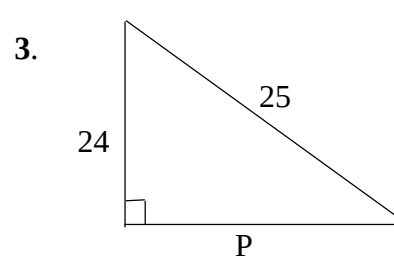
$$169 = y^2 + 144$$

$$169 - 144 = y^2$$

$$y^2 = 25$$

$$y = \sqrt{25}$$

$$y = \underline{5}$$



Solution

$$25^2 = p^2 + 24^2$$

$$625 = p^2 + 576$$

$$625 - 576 = p^2$$

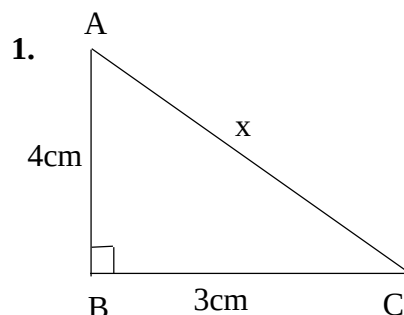
$$p^2 = 49$$

$$p = \sqrt{49}$$

$$p = \underline{7}$$



Calculate the length of the sides marked x in the right angle triangles.



Solution

$$AB^2 + BC^2 = AC^2$$

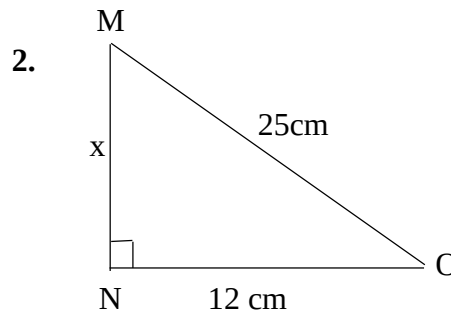
$$4^2 + 3^2 = x^2$$

$$16 + 9 = x^2$$

$$25 = x^2$$

$$x = \sqrt{25}$$

$$\underline{x = 5\text{cm}}$$



Solution

$$MN^2 + NO^2 = OM^2$$

$$x^2 + 12\text{cm}^2 = 25\text{cm}^2$$

$$x^2 + 144 = 625$$

$$x^2 = 625 - 144$$

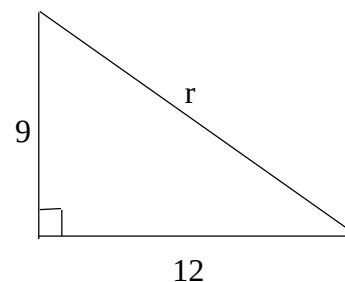
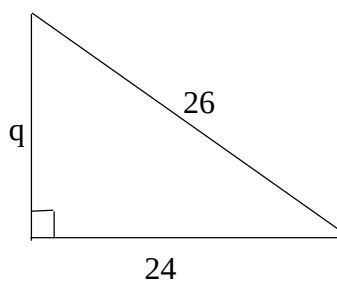
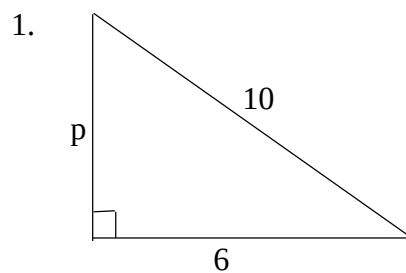
$$x^2 = 481$$

$$x = \sqrt{481}$$

$$\underline{x = 21.9317}$$

EXERCISE 1:

Calculate the length of the sides marked x in the right-angle triangles.



Exercise 2

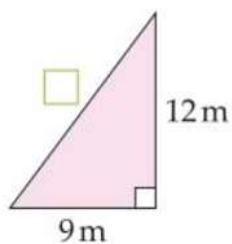


Pythagoras' Theorem

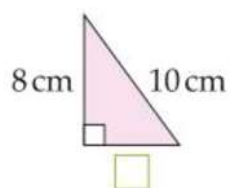
LO: Use the theorem to find a missing side in a right angled triangle.

Calculate the unknown lengths in these triangles.

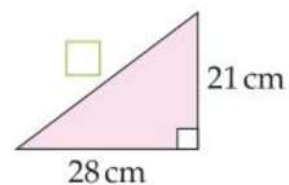
a



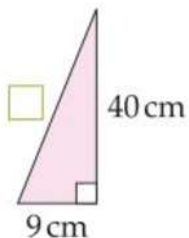
b



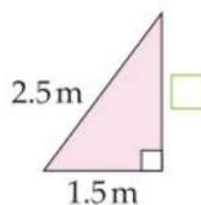
c



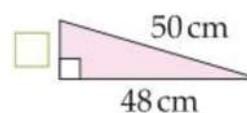
d



e



f





The Sierra Leone Teaching Service Commission



Exercise 3

1 A triangle has the same area as a square. If the area of the triangle 144cm^2 , what is the length of one side of the square.



The Sierra Leone Teaching Service Commission



UNIT 20.

CONGRUENCY TRIANGLES

Time:35 mins.

TEACHER'S GUIDE

Teachers to ensure the concepts of AAS, SAS, RAS are explained to students clearly.

STUDENT GUIDE

Students to draw the shape in order to see the similarities between two triangles.

LEARNING OUTCOMES

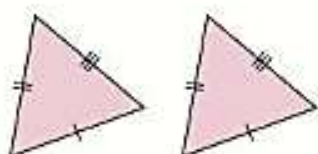
Pupils can use the properties of similar triangles to show that they are congruent.

Geometric Proof

LO: Use geometric facts to prove statements.

Two triangles are congruent if they satisfy one of four sets of conditions:

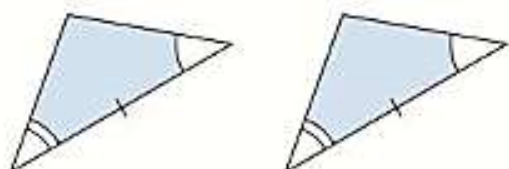
SSS: three sides the same



SAS: two sides and the included angle the same



ASA: two angles and the included side the same



RHS: right-angled triangles with hypotenuse and one other side the same.



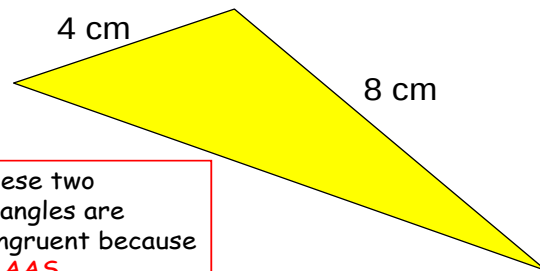
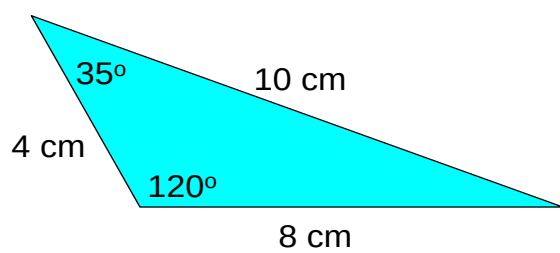
Congruency triangles are triangles that are similar in all respect. Such triangles are the same and identical dimensions. For triangles to be congruent, any one of the following conditions must be satisfied.

- All corresponding sides are equal (SSS)
- Two pairs of corresponding sides and the included angles are equal (SAS)
- Two pairs of corresponding angles and a pair of sides are equal (AAS)
- One pair of sides and a pair of corresponding angles are equal for right angles triangle (RHS).

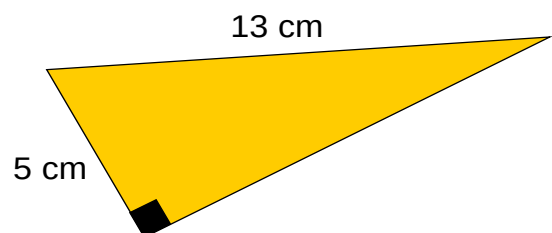
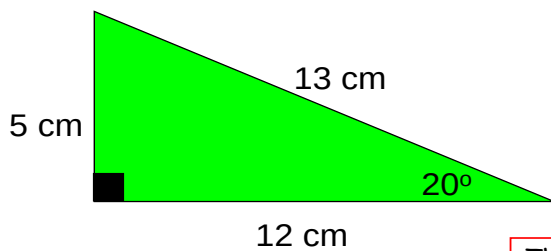


EXAMPLES

Congruency

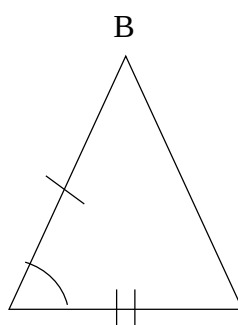
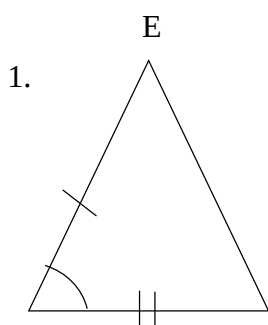


These two triangles are congruent because of **AAS**



These two triangles are congruent because of **RHS**

EXAMPLES:





The Sierra Leone Teaching Service Commission

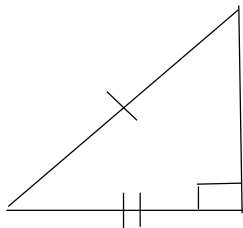


In

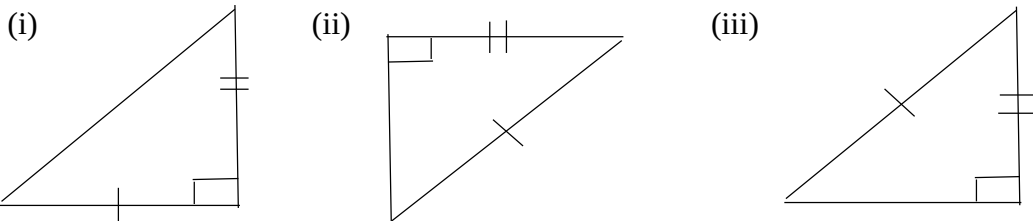
D	F	A	C
---	---	---	---

$\triangle DEF \triangle ABC$, two sides the same and the included angle of both triangles are equal.
 This condition abbreviated as (SAS).
 Therefore $\triangle DEF \equiv \triangle ABC$, SAS (side angle side).

2.



Which of the triangles below (i) (ii) and (iii) is/are congruent to the above given triangle.

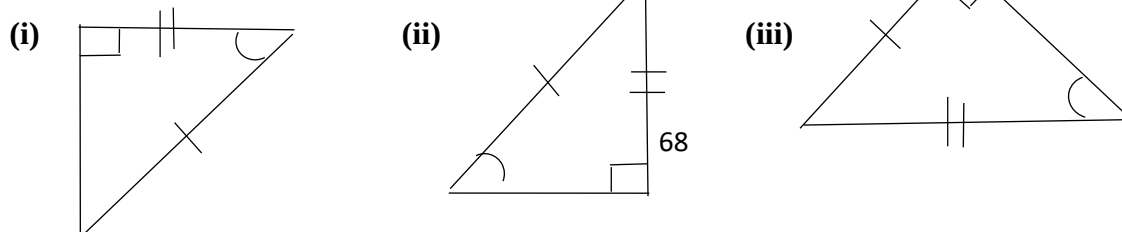
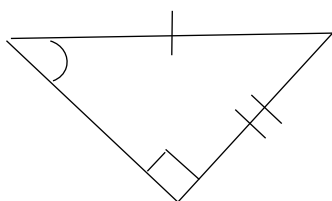


Answer: (ii) and (iii) are congruent.

EXERCISE

Which of the triangles (i) (ii) and (iii) is congruent to the given triangles?

a.





The Sierra Leone Teaching Service Commission



UNIT 21.

MONEY AND USE

Time: 35 mins.

TEACHER'S GUIDE

Teachers to use the column method addition, subtraction and subtraction.

STUDENT GUIDE

Students to use the column method for adding, subtracting and multiplying numbers.

LEARNING OUTCOMES

Pupils can carry out calculations involving money problems.

This involves the everyday uses of money.

Example I: A farmer collected 364 oranges from his farm on Monday, 2684 on Tuesday, 2560 on Wednesday, 3114 on Thursday, and 782 on Friday. He packs them in sacks of 300, to send them to Freetown by lorry

- a. How many oranges did he collected altogether?

Solution

Total number of oranges collected

$$= 364 + 2684 + 2560 + 3114 + 782 = \underline{9504}$$

- b. Number of oranges in one sack fill.

Solution

$$\frac{9504}{300} = 31 \text{ oranges in one sack}$$

300

Example II: if a dozen of onions cost Le 3,600 what is the cost of 9 dozens?

Solution

If 1 dozen = Le 3,600

Therefore 9 dozen = Le 3600 x 9

$$= \text{Le } 32,400$$

EXERCISE

1. Mr. Dauda has 13500 fruits trees in his garden. Two thirds of them are mango trees, the rest are orange trees. How many orange trees does he have?



The Sierra Leone Teaching Service Commission



2. Mr. Sankoh sold his house for one and half million Leones. He gave thirty-five thousand Leones to his son, fifty thousand to his wife and twenty-five thousand Leones to his girlfriend. How much did he keep for himself?
- 3a. The cost of a pen in a shop is Le 500 what is the cost of 12 pens bought from the same shop?
- b. If a customer pays Le 15,000, how many pens will be given to the customer?

UNIT 22. TRIGONOMETRY RATIOS

Time: 70 mins.

TEACHER'S GUIDE

Teacher to make clear understanding of opposite, adjacent and hypotenuse with one angle given inside the right-angle triangle.

STUDENT GUIDE

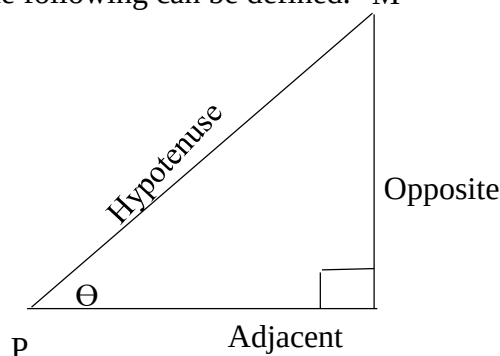
Students to copy examples set and complete the task set at the end of the unit.

LEARNING OUTCOMES

Pupils can calculate the missing sides and angles of the right-angled triangle.

Sin, cosine and tangent of angles

For a right-angled triangle, the following can be defined.



$$\text{Sine } \theta^\circ = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{MO}{MP}$$

$$\text{Cosine } \theta^\circ = \frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{PO}{PM}$$

$$\text{Tangent } \theta^\circ = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{MO}{PO}$$

NOTE:

- Opposite side of a right-angle triangle is the side facing the given angle beside the right angle.
- Hypotenuse is the longest side.
- Adjacent is the side joining the given angle and the right angle.



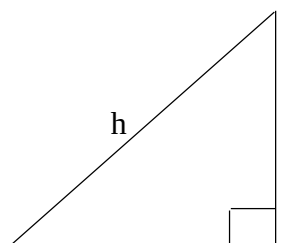
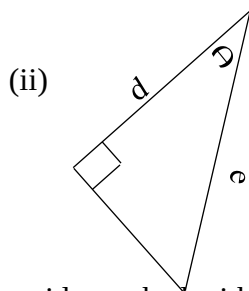
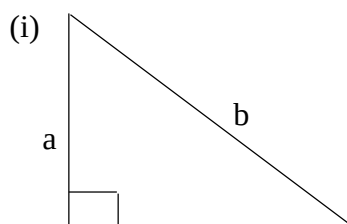
The Sierra Leone Teaching Service Commission



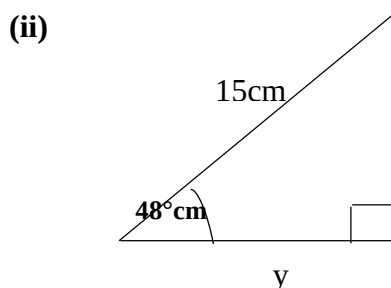
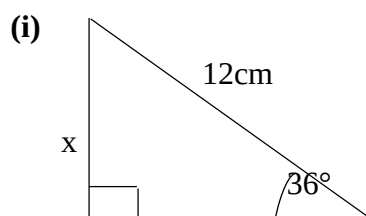
EXAMPLE 1:

Identify which of the letters in the triangle below are:

- a. Opposite
- b. Hypotenuse
- c. Adjacent

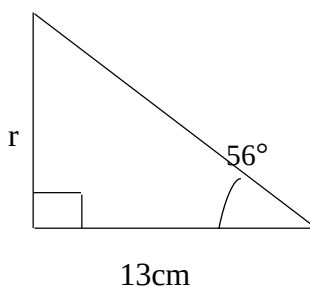


EXAMPLE 2: Find the values of the side marked with letters.



Solution

$$\frac{\sin 36}{1} = \frac{x}{12}$$
$$x = 12 \sin 36$$
$$x = 12 \times 0.5878$$
$$x = 7.0536$$
$$x = 7.1 \text{ cm (1dp)}$$



$$\frac{\tan 56}{1} = \frac{r}{13}$$
$$r = 13 \tan 56$$
$$r = 13 \times 1.483$$
$$r = 19.27$$
$$r = 19.3 \text{ cm (1dp)}$$

Solution

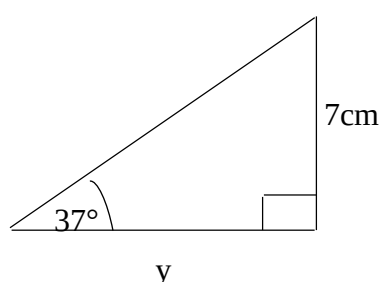
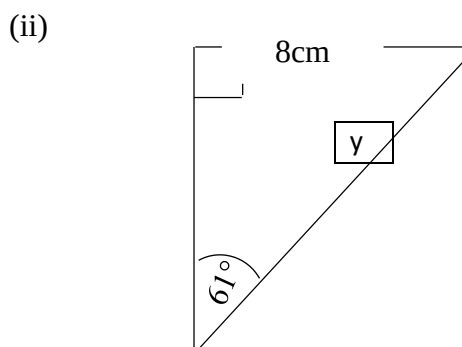
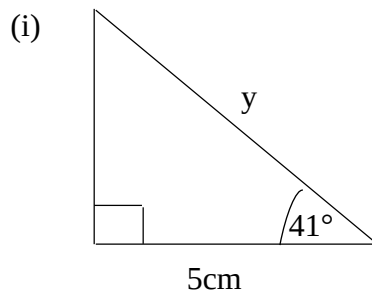
$$\frac{\cos 48^\circ}{1} = \frac{y}{15}$$
$$y = 15 \cos 48$$
$$y = 15 \times 0.6691$$
$$y = 4.668$$
$$y = 4.7 \text{ cm (1dp) (iii)}$$



The Sierra Leone Teaching Service Commission



EXERCISE (I) Find the length of the sides marked y in the diagram below, the triangle being right angled.



UNIT 23.

ALGEBRA

TEACHER'S GUIDE

Teacher to make the understanding of the key words: expression, simplify, term and like terms.

STUDENT GUIDE

Students to work independently in order to be able self-assess their work.

LEARNING OUTCOMES

Pupils can expand and factorize simple bracket.

Like Terms

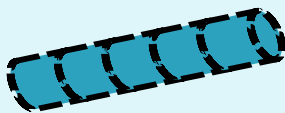


Try to write definitions for:

Expression, Simplify, Like Terms, Terms

- A **term** is part of an **expression**
- **Like terms** contain the same letter
- You can **simplify** an expression by collecting like terms

Like Terms



$5a$



$3a$

The total length = $5a + 3a = 8a$

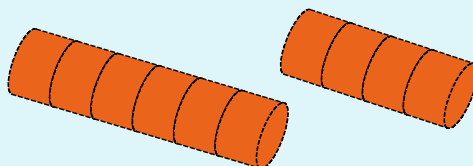
- **$5a$** and **$3a$** are like terms
- They can be combined, or collected together



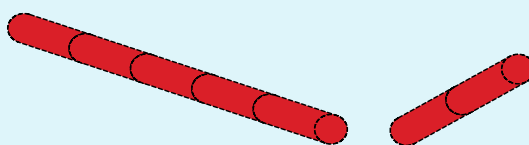
Exercises on Like Terms
simplify the expressions below



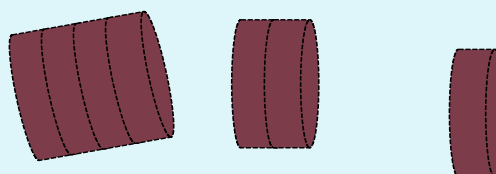
1a) $6x + 4x =$



1b) $5y + 2y =$



1c) $4z + 2z + z =$





Like Terms



Simplify each of the following,

a) $2n + 3n + n$

b) $4p + 2p - p$

c) $8q - 2q - 3q$

d) $5t + 3t + 4t$

e) $7r - 5r - r$

Expanding and factorizing brackets

1. Remove bracket and simplify

a. $5(a + 2) = 5a + 10$

b. $3(x - 4) = 3x - 12$

c. $5(a + 2) + 4(a + 1)$

Solution

$$5a + 10 + 4a + 4$$

$$5a + 4a + 10 + 4$$

$$9a + 14$$

d. $2(3x - y) + 3(x + 2)$

Solution

$$6x - 2y + 3x + 6y$$

$$6x + 3x = 2y + 6y$$

$$9x + 4y$$

e. $6(4x + y) - 7(3x + 5y)$

Solution

$$24x + 6y - 21x - 35y$$

$$24x - 21x + 6y - 35y$$

$$3x - 29y$$



The Sierra Leone Teaching Service Commission



2. Factorize the following

a. $ac - bc - ad + bd$

solution

$$c(a - b) - d(a - b)$$

$$(a - b)(c - d)$$

b. $xy - 4y - 5x + 20$

solution

$$y(x - 4) - 5(x - 4)$$

$$(x - 4)(y - 5)$$

EXERCISE 1

Expand the following brackets and simplify

(1) $3(x - 5y)$ (2.) $4x(2y + 3) - 3y(x + 8)$ (3.) $3(x + y - 2)$

(4.) $5p(3qr + s)$

UNIT 24.

LINEAR EQUATIONS

Time:35 mins.

TEACHER'S GUIDE

Teachers to engage students in working through the examples. Give them some time to produce some solutions to some of the questions.

STUDENTS GUIDE

Students to remember using the idea of collecting like terms.

LEARNING OUTCOMES

Pupils can solve problems involving linear equations.



The Sierra Leone Teaching Service Commission



Examples

Example 1

Solve $4x = 28$.

"4 times a number is 28"

The letter is multiplied by 4, so we must divide 28 by 4

$$\begin{aligned}\text{So } x &= 28 \div 4 \\ x &= 7\end{aligned}$$

Example 2

Solve $\frac{x}{6} = 3$

"A number divided by 6 is 3"

The letter is divided by 6, so we must multiply 3 by 6.

$$\begin{aligned}\text{So } x &= 3 \times 6 \\ x &= 18\end{aligned}$$

Exercises 1

1. $x + 7 = 18$
2. $g - 5 = 9$
3. $15 - h = 9$
4. $17 = b + 11$
5. $4y = 12$
6. $4a = 20$
7. $7a = 35$
8. $\frac{x}{12} = 4$

Answers:

1. $x = 11$
2. $g = 14$
3. $h = 6$
4. $b = 6$
5. $y = 3$
6. $a = 5$
7. $n = 40$
8. $x = 48$

Solve the following linear equations with brackets.

a. $13 - 6a = 1$

Solution

$$-6a = 1 - 13$$

$$-6a = -13$$

b. $2(x + 5) = 18$

Solution

$$2x + 10 = 18$$

$$2x = 18 - 10$$

c. $5(a + 2) = 4(a - 1)$

Solution

$$5a + 10 = 4a - 4$$

$$5a - 4a = -4 - 10$$

d. $2(x + 3) = 7$

Solution

$$2x + 6 = 7$$

$$2x = 7 - 6$$



The Sierra Leone Teaching Service Commission



$$\begin{aligned} -6a &= -12 \\ -6 & \quad 6 \\ \underline{a} &= \underline{2} \end{aligned}$$

$$\begin{aligned} \frac{2x}{2} &= \frac{8}{2} \\ \underline{x} &= \underline{4} \end{aligned}$$

$$\underline{a = -14}$$

$$\frac{2x}{2} = \frac{1}{2}$$

EXERCISE 2

- a. $5(x - 4) = 4(x + 1)$
- b. $2(y - 2) + 3(y - 7) = 0$
- c. $15 = 3(x - 3)$

Unit 25

CHANGE OF SUBJECT AND SUBSTITUTION

Time: 70 mins.

TEACHERS GUIDE

Teacher s to go through the examples slowly for pupils to understand change of subject.

STUDENT GUIDE

Students to practice the examples and complete the exercises set.

LEARNING OUTCOMES

Pupils can make a variable the subject of the formula and then substitute a number to get the result.

Example 1

**Change the subject in the following
to make a the subject**

$$v = u + at$$

$$v - u = at$$

$$\frac{v - u}{t} = a$$



Question 2

Change the subject in the following
to make p the subject

$$pk + q = h$$

$$pk = h - q$$

$$p = \frac{h - q}{k}$$

1. Make x the subject of the relation.

$$\frac{x-1}{m} + \frac{y}{n} = 2$$

2. Given that $n = \frac{mk}{ec}$, make c the subject

solution

$$\frac{n(x-1) + my}{mn} = 2$$

$$\frac{nx - n + my}{mn} = 2$$

$$nx + n + my = 2mn$$

$$nx = 2mn - my - n$$

$$\frac{x = 2mn - my - n}{n}$$

3. Make m the subject of the relation.

$$h = \frac{mt}{d+p}$$

solution

$$\frac{h}{t} = \frac{m}{d+p}$$

Solution

$$n = \frac{mk}{ec}$$

$$nec = mk$$

$$\frac{nec}{ne} = \frac{mk}{ne}$$

$$c = \frac{mk}{ne}$$



$$mt = h(d + p)$$

$$\frac{mt}{t} = h \frac{(d + p)}{t}$$

$$m = h \frac{(d + p)}{t}$$

EXERCISE

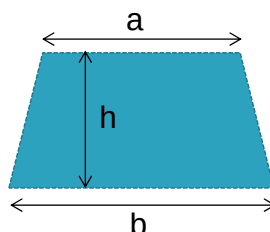
1. Make x the subject of the relation $2x + 4y = 10$
2. Make x the subject: $x + y = 9$
3. Make t the subject of the relation $k = \frac{t - 3}{t - 9}$

SUBSTITUTION

SUBSTITUTION

The formula for the area of a trapezium is:

$$A = \frac{(a + b)h}{2}$$



Find the area of a trapezium when $a = 5$, $b = 9$ and $h = 3$:

$$A = \frac{(a + b)h}{2}$$

$$A = \frac{(5 + 9)3}{2}$$

$$A = \frac{14 \times 3}{2} \quad A = \frac{42}{2} \quad A = 21$$



The Sierra Leone Teaching Service Commission



EXAMPLES

1. If $a = 5$, $b = 4$ and $c = 3$, evaluate the following.

a. $3a + 6$ (b) $5 + ab$ (c) $a + bc$ (d) $5ab + 6bc$ (e) $4a + 6$

Solution

(a) $3a + 6$	(b) $5 + ab$	(c) $a + bc$	(d) $5ab + 6bc$	(e) $4a + 6$
$= 4(5) + 6$	$= 5 + 5 \times 4$	$= 5 + 4 \times 3$	$= 5(5)(4) + 6(4)(3)$	$= 4(a) + 6$
$= 15 + 6$	$= 5 + 20$	$= 5 + 12$	$= 100 + 72$	$= 4 \times 5 + 6$
$= 21$	$= 25$	$= 17$	$= 172$	$= 20 + 6$
				$= 26$

EXERCISE

1. Evaluate the following expressions given that

$a = 5$, $b = 2$, $c = 3$, $u = 4$ and $v = -3$

1. $(a \times c)$ (2) $(u - v)$ (3) $\frac{3+a}{3} + \frac{u+c}{3}$ (4) $(a - b)(u - v)$

UNIT 26.

QUADRATIC EQUATIONS

Time: 70 mins.

TEACHER'S GUIDE

Teacher to start with quadratics equations with the coefficient of x^2 term equal to 1. Teacher should introduce the key words product and sum. that the sum of the 2 factors give coefficient of the middle term and

STUDENT GUIDE

Students to work independently with support of the teacher.

LEARNING OUTCOMES

Pupils can solve problems with quadratic equations.



The Sierra Leone Teaching Service Commission



A quadratic expression is one obtained from the product of two linear expression in the same variable. Solve the following quadratic equations.

1. $x^2 - 8x + 16 = 0$

solution

$$\begin{aligned} x^2 - 4x - 4x + 16 &= 0 \\ x(x - 4) - 4(x - 4) &= 0 \\ \text{either} \\ x - 4 &= 0 \text{ or } x - 4 = 0 \\ x &= 4 \text{ or } x = 4 \end{aligned}$$

2. $x^2 + 9x + 20 = 0$

Solution

$$\begin{aligned} x^2 + 4x + 5x + 20 &= 0 \\ x(x + 4) + 5(x + 4) &= 0 \\ (x + 4)(x + 5) &= 0 \\ \text{either} \\ x + 4 &= 0 \text{ or } x + 5 = 0 \\ x &= -4 \text{ or } x = -5 \end{aligned}$$

3. $3x^2 - 2x - 8 = 0$

Solution

$$\begin{aligned} 3x^2 - 2x - 8 &= 0 \\ 3x^2 + 4x - 6x - 8 &= 0 \\ x(3x + 4) - 2(3x + 4) &= 0 \\ (3x + 4)(x - 2) &= 0 \\ \text{Either } 3x + 4 &= 0 \text{ or } x - 2 = 0 \\ x &= \frac{-4}{3} \text{ or } x = 2 \end{aligned}$$

4. $x^2 + 2x - 8 = 0$

Solution

$$\begin{aligned} x^2 - 2x + 4x - 8 &= 0 \\ x(x - 2) + 4(x - 2) &= 0 \\ (x - 2)(x + 4) &= 0 \\ \text{either } x - 2 &= 0 \text{ or } x + 4 = 0 \\ x &= 2 \text{ or } x = -4 \end{aligned}$$

EXERCISE

1. $a^2 + 7a + 12 = 0$
2. $x^2 - 10x + 21 = 0$
3. $y^2 + 8y + 7 = 0$

UNIT 27

INEQUALITY

Time: 35mins

Teacher's guide

Teacher to advise students to use a ruler and pencil to draw straight lines.

Student guide



The Sierra Leone Teaching Service Commission



Students to work through the exercises and ensure you draw a line graph for every question.

LEARNING OUTCOMES

Pupils can solve problems involving inequalities.

NOTATIONS

$<$ $\longrightarrow$ Less than

$>$ $\longrightarrow$ Greater than

$\leq$ $\longrightarrow$ Less than or equal to

$\geq$ $\longrightarrow$ Greater than or equal to

Inequalities on number lines

The symbols $<$, $\leq$, $>$ and $\geq$ are used to express inequalities (things that are **not** equal).

$x < 3$ means $2 < x$ means

$x \leq 5$ means $-1 \leq x$ means

$x > 4$ means $10 > x$ means

$x \geq -3$ means $6 \geq x$ means

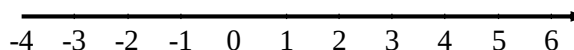
$2 < x < 7$ means

$1 \leq x \leq 9$ means

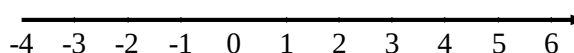
It is possible to represent inequalities on a number line.

Examples

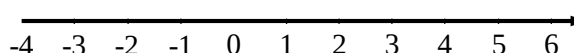
1. The inequality $x < 5$ can be represented as



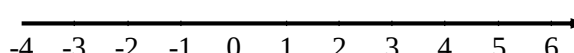
2. The inequality $x \geq 2$ can be represented as



3. The inequality $1 < x < 4$ can be represented as

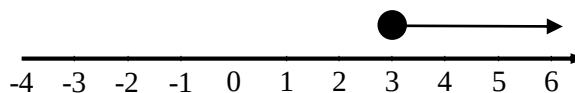


4. The inequality $-2 \leq x \leq 3$ can be represented as

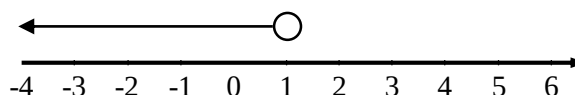




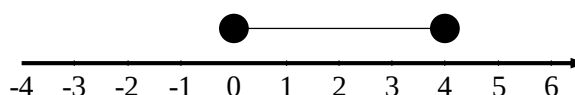
5. The inequality represented in the diagram is



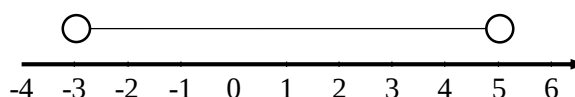
6. The inequality represented in the diagram is



7. The inequality represented in the diagram is



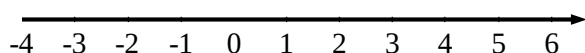
8. The inequality represented in the diagram is



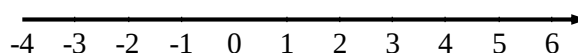
Exercise

1. Represent each inequality on the number line.

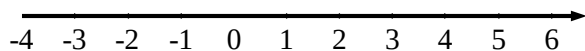
(a) $x < 4$



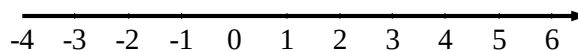
(b) $x \geq 1$



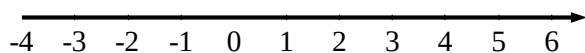
(c) $x > 0$



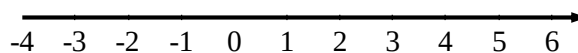
(d) $x \leq 3$



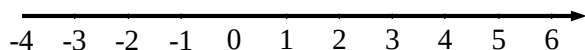
(e) $x \geq -2$



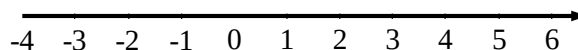
(f) $x < 2$



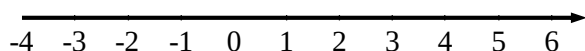
(g) $1 < x$



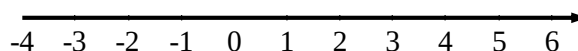
(h) $4 \geq x$



(i) $1 < x < 5$



(j) $0 \leq x \leq 3$

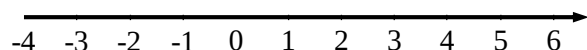


(k) $-3 \leq x \leq 2$

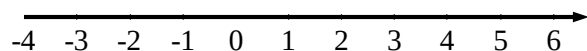
(l) $-1 < x < 4$



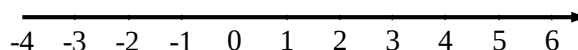
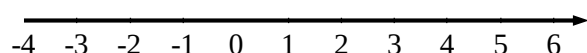
The Sierra Leone Teaching Service Commission



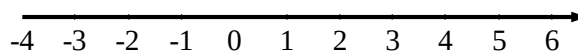
(m) $2 < x \leq 5$



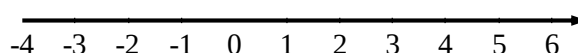
(o) $-2 \leq x < 1$



(n) $0 \leq x < 3$

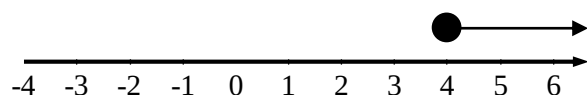


(p) $-3 < x \leq -1$

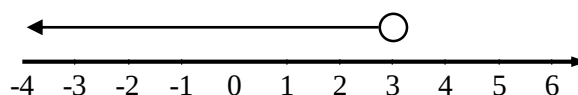


2. Write down the inequality represented in each part.

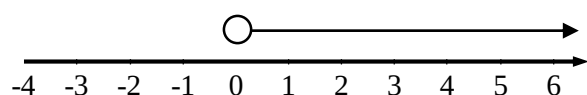
(a)



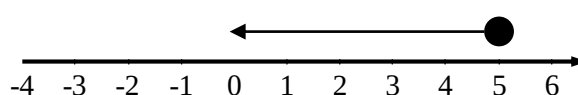
(b)



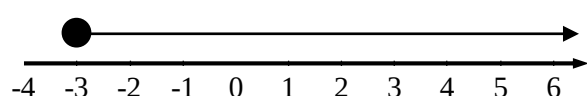
(c)



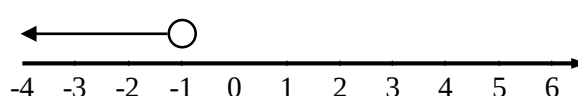
(d)



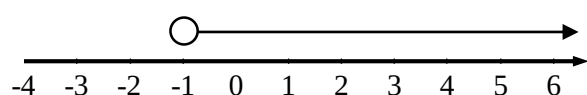
(e)



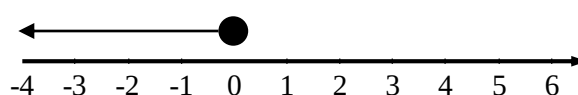
(f)



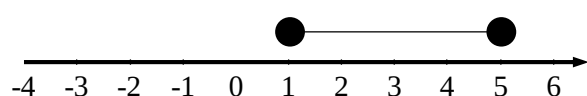
(g)



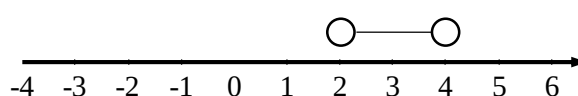
(h)



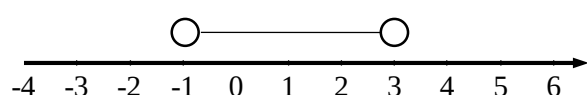
(i)



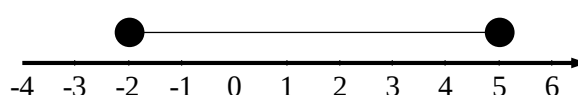
(j)



(k)

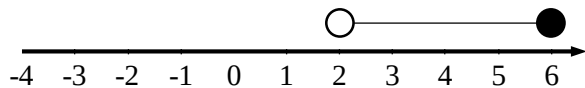


(l)

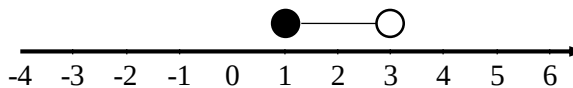




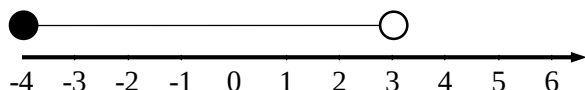
(m)



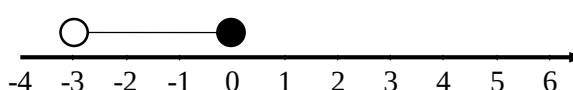
(n)



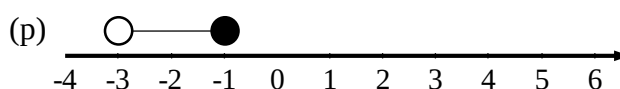
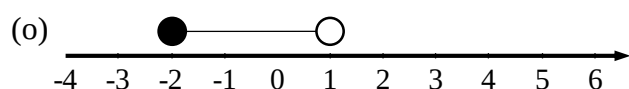
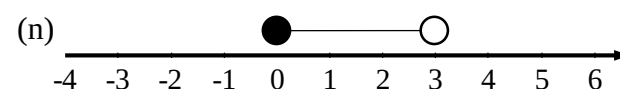
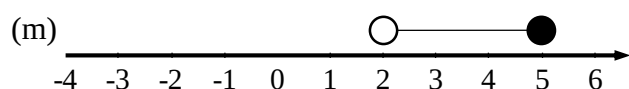
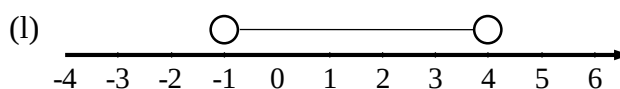
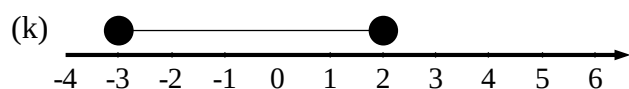
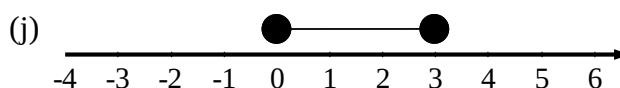
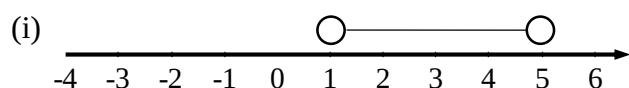
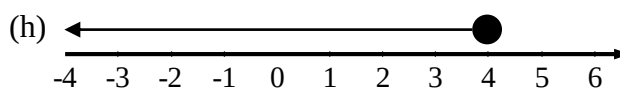
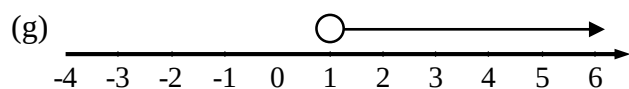
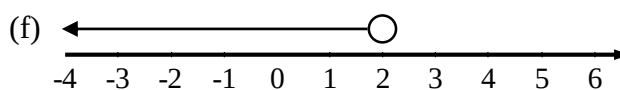
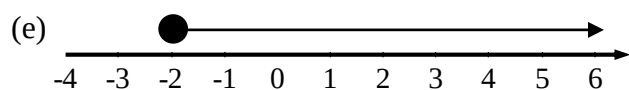
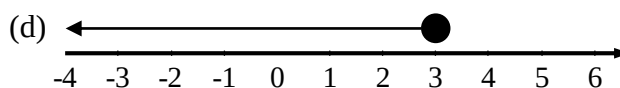
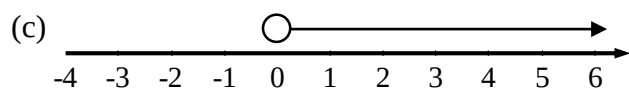
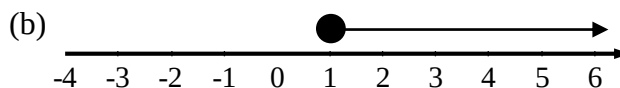
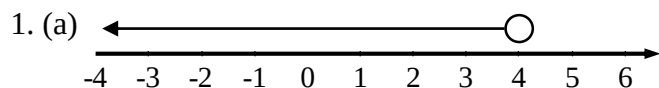
(o)



(p)



Answers



2. (a) $x \geq 4$

(b) $x < 3$

(c) $x > 0$

(d) $x \leq 5$

(e) $x \geq -3$

(f) $x < -1$



The Sierra Leone Teaching Service Commission



(g) $x > -1$

(h) $x \leq 0$

(i) $1 \leq x \leq 5$

(j) $2 < x < 4$

(k) $-1 < x < 3$

(l) $-2 \leq x \leq 5$

(m) $2 < x \leq 6$

(n) $1 \leq x < 3$

(o) $-4 \leq x < 3$

(p) $-3 < x \leq 0$

UNIT 28

COLLECTION AND PRESENTATION OF DATA

Time: 70 mins.

TEACHER'S GUIDE

Teacher to guide through on how to draw bar graphs, histogram and pie charts.

STUDENT GUIDE

Pupils to practice representing data on graphs.

LEARNING OUTCOMES

Pupils can represent data on a graph.

NOTE:

1. Data can be organised and clearly presented in many ways. These include the following.
 - a. Rank order list – where data is placed in numerical order from highest to lowest.
 - b. Frequency table – where the number of times a particular event happens is recorded in a table.
(Frequency means the number of times something happens.)
2. Data can also be represented in a graph, or picture. Examples include the following.



The Sierra Leone Teaching Service Commission



- a. **Pictogram** – where pictures or drawings represent data.
- b. **Bar chart** – where the length or height of a bar is proportional to the data
- c. **Pie chart** – where the size of the sector of a circle is proportional to the data.

Frequency Table

Frequency means the number of times something happens. For example, three students got grade A.

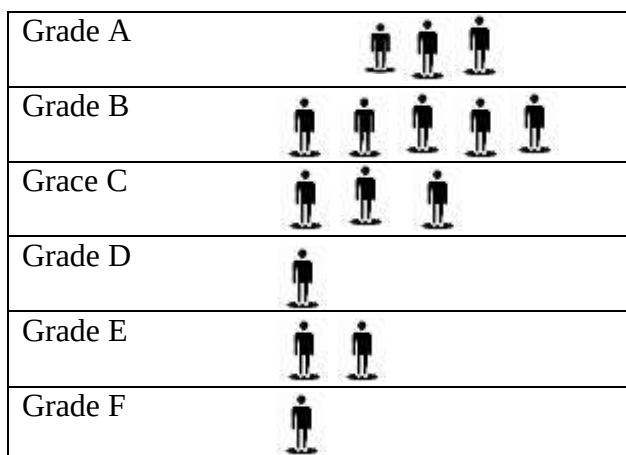
Grade	A	B	C	D	E	F
Frequency	3	5	3	1	2	1

- The frequency of grade A is three.

PICTOGRAM

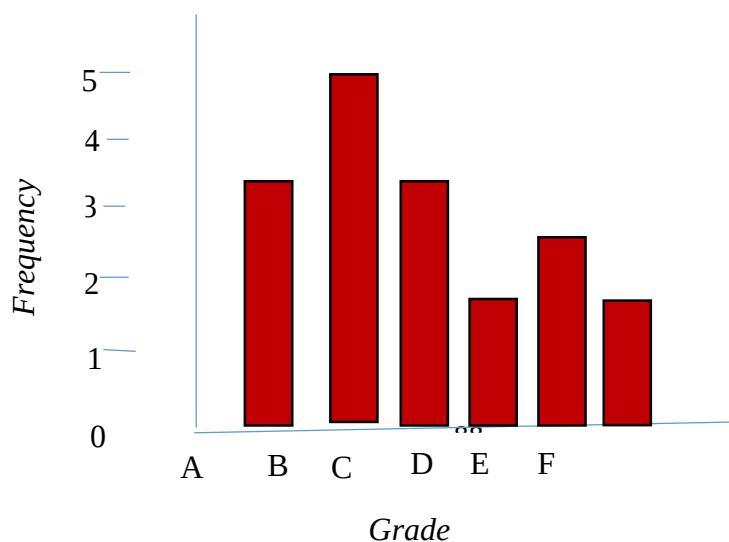
A pictogram uses pictures or drawings to give a quick and easy meaning to statistical data.

In the pictogram below each pin figure represents a student who gets the grade shown.



BAR CHART

A bar chart is very like pictogram. The number of students who get each grade is represented by a bar instead of a picture. The bars have the same width and usually have equal spaces between them. The height of each bar below represents the frequency of that grade.





The Sierra Leone Teaching Service Commission



PIE CHART

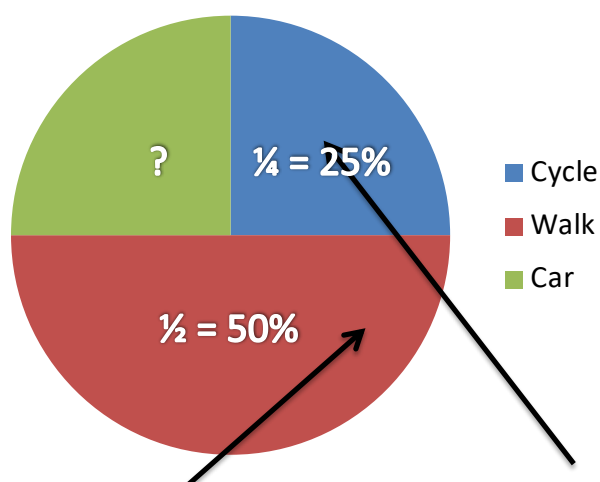
A pie chart is a graph in the shape of a circular pie. In the circle the total of student (15) make up the whole pie of 360° . Each piece of the pie is a sector of the circle. The size of each sector represents the number of students who get the grade shown in that sector.

Here is a pie chart.

1. A pie chart is a visual chart used to display data.
2. It is usually circular and is split up into different sections (like a pie).
3. Pie charts are normally in either percentages (25%) or fractions ($\frac{1}{2}$).

This is a pie chart showing how 40 children travelled to school.

Now we look at how many people were surveyed.



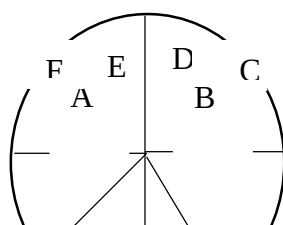
How many children...
Cycled?
Walked?
Were drove?

How do we know how many children did what?

What fraction is this piece of the whole circle (pie)?

How about this piece?

Grade	A	B	C	D	E	F	Total
Frequency	3	5	3	1	2	1	15
Angle at centre	72°	120°	72°	24°	48°	24°	360°





The Sierra Leone Teaching Service Commission



The angles are calculated on the basis of simple ratio for example

$$\text{Angle for grade A} = \frac{360^\circ}{15} \times 3 = 72^\circ$$

Or $\frac{3}{15} \times 360 = 72^\circ$

$$\text{Grade B} = \frac{5}{15} \times 360 = 120^\circ$$

Handling Data

LO: Construct pie charts.

Ice-cream Sales	
Vanilla	13
Banana	22
Chocolate	28
Strawberry	57
Total	120

The information in the pie chart shows sales of 120 ice-creams sold from an ice-cream van one Saturday afternoon in the summer.

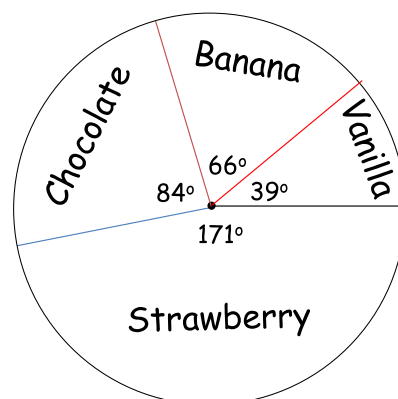
Calculate the number of each type sold.

$$\text{Vanilla} = \frac{39}{360} \times 120 = 13$$

$$\text{Banana} = \frac{66}{360} \times 120 = 22$$

$$\text{Chocolate} = \frac{84}{360} \times 120 = 28$$

$$\text{Strawberry} = \frac{171}{360} \times 120 = 57$$





The Sierra Leone Teaching Service Commission



Time: 35 mins

Averages

Finding Mode, Median, Mean and Range

Example:

Mode= The value that occurs the most

Median= The middle value when the values are put in order from smallest to biggest

Mean=
$$\frac{\text{Sum of all values in the set}}{\text{Number of values in the set}}$$

(sometimes referred to as the “average”)

Range= Biggest value- smallest value **Note:**

Putting the values in order from smallest to biggest will make it easier to work out your averages!

4, 9, 4, 6, 10, 6, 3, 5, 4, 9

Arranged in order

3, 4, 4, 4, 5, 6, 6, 9, 9, 10

Mode= 4

Median= 3, 4, 4, 4, 5, 6, 6, 9, 9, 10

If there are 2 numbers left in the middle, add them up and divide by 2!

So median= $(5 + 6) / 2 = 5.5$

MEAN = $(3+4+4+4+5+6+9+9+10) \div 10$

RANGE = $10 - 3 = 7$

EXERCISES



The Sierra Leone Teaching Service Commission



- 1) The set of values shows the number of goals scored by a school football team in their first 10 matches: 2, 4, 1, 0, 2, 3, 2, 6, 2, 4

Find: a) Mode =

c) Median =

a) Mean =

d) Range

Find the mode, mean, median and range for each set of values given:

- 2) 7, 2, 5, 3, 8, 5, 5

Mode =

Median =

Mean =

Range =

- 3) 1, 7, 2, 5, 3, 1, 1, 4

Mode =

Median =

Mean =

Range =

- 4) 6, 9, 9, 1, 5

Mode =

Median =

Mean =

Range =

Unit 29

PROBABILITY

Time: 70 mins.

Teacher's guide

Teacher to carefully explain the probability scale ranging from zero to one (0 - 1).

STUDENT GUIDE

Pupils to copy the probability scale in their exercise books.

LEARNING OUTCOMES

Pupils can solve problems involving simple probability problems.

The probability that an event will happen is a number between 0 and 1.

The probability for an event which is **CERTAIN = 1**

The probability for an event which is **IMPOSSIBLE = 0**



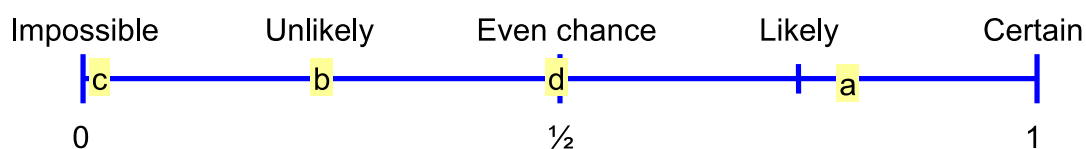
Probability can be expressed as a fraction, a decimal or a percentage.

The probability scale

Probability is a measure of how likely or unlikely an event is to occur.

Usually written as fractions, but can be written in any form **equivalent** to that fraction. Eg $\frac{3}{4} = 0.75 = 75\%$

Can be anywhere between 0 (impossible) and 1 (certain):



a) an event with a probability of $\frac{3}{4}$ would be described as **likely**

b) an event with a probability of $\frac{1}{4}$ would be described as **unlikely**

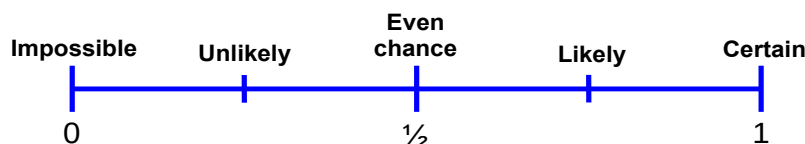
c) an event with a probability of 0.01 would be described as **very unlikely**

d) an event with a probability of $\frac{1}{2}$ would be described as **even chance**

**Key words:**

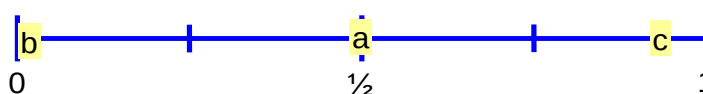
Even chance
Likely
Impossible
Certain
Unlikely

1. Complete this probability scale using the key words given



2. Label the events described below on the probability scale:

- a) The chance of getting an even number when rolling a dice
- b) The chance of winning the National Lottery
- c) The chance of rain in March



3. Describe an event that you think has a probability of:

- a) 0.3 _____
- b) 1 _____
- c) 0.8 _____

Probability of an event

The probability of an event occurring is defined as:

$$\text{Probability} = \frac{\text{Number of desired outcomes}}{\text{Number of possible outcomes}}$$

Eg the probability of rolling an even number on a fair dice

Desired outcomes are even numbers 2, 4 and 6 (**three** of them)

Possible outcomes are the numbers 1 to 6 (**six** of them)

$$\text{so Probability} = \frac{3}{6}, \text{ which simplifies to } \frac{1}{2}$$

Eg Mr Conteh puts all the pupils' names in a hat and picks one at random.

What is the probability he picks a SS2 student?

Desired outcomes are any SS2 picked (**15** of them)

Possible outcomes are any pupil (**130** of them)

$$\text{so Probability} = \frac{15}{130}, \text{ which simplifies to } \frac{3}{26}$$

FORM	Pupils
JSS1	22
JSS2	16
JSS3	28
SS1	29
SS2	15
SS3	20



Exercises

Probability of an event

1. Bob is picking randomly from a bag containing tiles numbered 1 to 10.

Write down the probability that the number he picks is:

- a) $\frac{7}{10}$ b) 4 less $\frac{4}{10} = \frac{2}{5}$ c) Odd $\frac{5}{10} = \frac{1}{2}$ d) A multiple of 3

2. A survey is conducted of pupils' favorite team:

Team	Spurs	Man Utd	Liverpool	Arsenal
Pupils	12	8	4	6

John picks a pupil at random to ask more questions.

Write down the probability that the pupil he picks supports:

- a) Liverpool b) A London team c) Not Liverpool $\frac{13}{15}$

$$P(\text{Liverpool}) = \frac{4}{30} = \frac{2}{15} = 0.4$$

$$P(\text{London Team} = \text{Spurs} + \text{Arsenal}) = \frac{18}{30} = \frac{3}{5} = 0.6$$

$$\frac{4}{30} = \frac{2}{15}$$

$$\frac{18}{30} = \frac{3}{5}$$

3. A bag contains 20 coloured balls, some red and some blue.

Abu knows that the probability of picking a red ball is $\frac{2}{5}$.

How many red balls are there?



Probability using tables

Eg in a game, two fair dice are rolled, and a score is found by multiplying the numbers obtained together.

- Show the possible outcomes in the table below
- Use your completed table to find the probability of getting a score of 12
- Use the table to find the probability of getting a score of 23 or more

a)

		Dice A					
Dice B		1	2	3	4	5	6
	1	1	2	3	4	5	6
	2	2	4	6	8	10	12
	3	3	6	9	12	15	18
	4	4	8	12	16	20	24
	5	5	10	15	20	25	30
	6	6	12	18	24	30	36

$$\text{Probability} = \frac{\text{Number of desired outcomes}}{\text{Number of possible outcomes}}$$

- b) 4 outcomes out of 36
give a score of 12

$$\text{Probability} = \frac{4}{36} = \frac{1}{9}$$

- c) 6 outcomes out of 36
give a score of 23+

$$\text{Probability} = \frac{6}{36} = \frac{1}{6}$$

Probability

Dice



1. A fair dice is rolled.

What is the probability of rolling:

(a) 6

(b) 3

(c) 2 or 3

(d) even number

(e) number greater than 2

2. A fair dice is rolled.

What is the probability of rolling: (a) 1 (b) 2, 3 or 4



The Sierra Leone Teaching Service Commission



(c) 3 or 5 (d) number less than 2

3. A fair dice is rolled.

What is the probability of rolling: (a) 1 or 3 (b) 2,3,4 or 5



Answers

Dice



1. A fair dice is rolled.
What is the probability of rolling: (a) 6 = $\frac{1}{6}$ (b) 3 = $\frac{1}{6}$
(c) 2 or 3 = $\frac{1}{3}$ (d) even number = $\frac{1}{2}$
(e) number greater than 2 = $\frac{2}{3}$
2. A fair dice is rolled.
What is the probability of rolling?
(a) 1 = $\frac{1}{6}$ (b) 2, 3 or 4 = $\frac{1}{2}$
(c) 3 or 5 = $\frac{1}{3}$ (d) number less than 2 = $\frac{1}{6}$
3. A fair dice is rolled.
What is the probability of rolling?
(a) 1 or 3 = $\frac{1}{3}$ (b) 2, 3, 4 or 5 = $\frac{2}{3}$
(c) 7 = 0 (d) number less than 10 = 1.

Unit 30

The use of the internet and computers to aid students learning.

There are several mathematics resources available for students online but could be expensive for some of our students who live in the remote areas of Sierra Leone. For those students who can access the internet could use the following websites like [bbc bitesize](#), [Mymaths](#), [myimaths](#).



ANSWERS

UNIT 10

- 1) 4
2) 5, 6

UNIT 11 EX1

A)280 B) 25 C

EXERCISE 2

a)8 b)15 c)140

Unit 12 Ex 1

i)96 ii)182 iii)81

exercise 2

1)530 2)864 3)472 4)805

unit 14

\$640, 78101.4 Yen, Euro 402.50, 40061 Rupees, 5315Rand, 39.5 Australian dollars

Unit 15

1)8.8cm 2) 28cm² 3i)10.95cm 3ii) 5.9cm 4) 113cm² 5)6.8cm
6i)45cm² 6ii)48cm² 7) 7.25cm

UNIT 16

1)28.27cm² 2) 78.54cm 3)153.9cm²

Unit 17

1)1539.4cm³ 2)1120cm³ 3)1.3cm

unit 19

Exercise 1

1)8 2)10 3)15

Exercise 2

a)15m b)6cm c)35cm d)41cm e)2cm f)14

Exercise 3

12cm

Unit 20

ii and iii

unit 21

1)4500 Orange trees
2)Le 1,390,000
3a) Le 6000 (3b) 30 pens

unit 22

h) 6.6 ii)9 .1cm iii) 9.3cm

Unit 23

1) 3x -15y

UNIT 24

a) x =24



Teaching Service Commission Sierra Leone (TSC-SL)



2) $5xy + 12x - 24y$

3) $3x + 3y - 6$

4) $15pqr + 5ps$

b) $y = 17$

c) $x = 8$

UNIT 25

1) $x = 5 - 2y$

3) $t = \frac{9k - 3}{k - 1}$

2) $x = 9 - y$

$k - 1$

UNIT 26

1) $a = -3$ or -4

2) $x = 3$ or 4

3) $y = -1$ or -7